

Preparation of biogenic gas vesicle nanostructures for use as contrast agents for ultrasound and MRI

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Gas vesicles (GVs) are a unique class of gas-filled protein nanostructures that are detectable at subnanomolar concentrations and whose physical properties allow them to serve as highly sensitive imaging agents for ultrasound and MRI. Here we provide a protocol for isolating GV from native and heterologous host organisms, functionalizing these nanostructures with moieties for targeting and fluorescence, characterizing their biophysical properties and imaging them using ultrasound and MRI. GV can be isolated from natural cyanobacterial and haloarchaeal host organisms or from *Escherichia coli* expressing a heterologous GV gene cluster and purified using buoyancy-assisted techniques. They can then be modified by replacing surface-bound proteins with engineered, heterologously expressed variants or through chemical conjugation, resulting in altered mechanical, surface and targeting properties. Pressurized absorbance spectroscopy is used to characterize their mechanical properties, whereas dynamic light scattering (DLS) and transmission electron microscopy (TEM) are used to determine nanoparticle size and morphology, respectively. GV can then be imaged with ultrasound *in vitro* and *in vivo* using pulse sequences optimized for their detection versus background. They can also be imaged with hyperpolarized xenon MRI using chemical exchange saturation transfer between GV-bound and dissolved xenon—a technique currently implemented *in vitro*. Taking 3–8 d to prepare, these genetically encodable nanostructures enable multimodal, noninvasive biological imaging with high sensitivity and potential for molecular targeting.

INTRODUCTION

The study of biological function and disease, and the development of clinical diagnostics require technologies for noninvasive imaging of cells and molecules in intact organisms. Ultrasound and MRI are two widely used noninvasive imaging modalities that provide high spatial and temporal resolution but are currently restricted in their molecular imaging capabilities because of limited repertoires of nanoscale contrast agents. For ultrasound, conventional imaging agents based on microbubbles—micron-sized particles of trapped gas stabilized by a lipid or protein shell—are limited by their size and physical instability to labeling the vasculature and endovascular targets^{1,2}. Conventional MRI contrast agents based on superparamagnetic iron oxides^{3,4} or lanthanide chelates⁵ are limited by their potential toxicity and the requirement that they be present at relatively high concentrations (typically in the micromolar range) for detection by MRI.

Development of the protocol

To develop biomolecular imaging agents addressing the aforementioned limitations, we recently introduced a new class of molecular imaging agents for ultrasound and MRI based on GV—genetically encoded gas-filled protein nanostructures from buoyant photosynthetic microorganisms^{6,7}. GV comprise a protein shell with a hydrodynamic diameter of ~200 nm and thickness of ~2 nm enclosing a hollow gas-filled interior^{6,7}. Gas dissolved in the surrounding media partitions freely in and out of this nanoscale compartment, whereas liquid water is excluded by the shell's hydrophobic inner surface. Microbes such as the cyanobacterium *Anabaena flos-aquae* (Ana) and the archaeon *Halobacterium salinarum* (Halo) form GV as a means to regulate cellular buoyancy for optimal access to light and nutrients. In these organisms,

GVs are encoded by operons of 8–14 genes, which include the primary structural proteins GvpA and GvpC and several minor constituents and chaperones^{6,7}.

We recently discovered that GV purified from cyanobacteria and haloarchaea scatter sound waves and thereby produce ultrasound contrast at subnanomolar concentrations⁸. Furthermore, we found that GV's unique physical properties enable them to produce harmonic ultrasound signals useful for contrast specificity *in vitro* and *in vivo*^{9,10}. We also showed that GV from different species can be imaged in multiplex on the basis of their differential responses to acoustic pressure, and that conditional GV clustering leads to contrast enhancement, allowing them to be used as molecular sensors⁸. In parallel, we demonstrated that GV produce contrast in hyperpolarized xenon MRI¹¹, an emerging form of imaging that takes advantage of nonequilibrium spin polarization to increase molecular sensitivity by factors of up to 10⁵. We discovered that atoms of hyperpolarized ¹²⁹Xe dissolved in aqueous media exchange in and out of the GV interior, where their distinct chemical shift enables amplified MRI contrast to be produced using chemical exchange saturation transfer (CEST) pulse sequences. Using this hyperpolarized CEST (HyperCEST) technique allows GV to be detected at picomolar concentrations¹¹. In addition, the distinct chemical shifts of GV in different microbial species enable multiplexed imaging. HyperCEST is currently performed *in vitro* but is advancing toward *in vivo* use^{12,13}.

One particularly promising aspect of GV is the possibility of engineering their physical and biochemical properties at the genetic level. To take advantage of this possibility, we recently developed a convenient genetic engineering platform in which a single target protein, GvpC, on the surface of GV purified from

Ana cells is replaced by heterologously expressed engineered versions¹⁴. This strategy allows the modification of GV mechanical properties, including their harmonic response to sound waves and collapse under pressure for enhanced contrast and multiplexed imaging. In addition, genetic modification enables surface display of peptides such as lysine-rich protein to modify surface charge, CD47 to minimize uptake by macrophages, polyarginine for cellular internalization and RGD for targeting to integrin receptors on tumor cells¹⁴.

Overview of the procedure

Here, we present a detailed optimized protocol to express and purify GVs (Step 1A–C) and quantify (Steps 2 and 3), functionalize and genetically engineer them (Steps 4–45), characterize their size and shape (Step 46A–C), and use them as contrast agents for noninvasive imaging applications (Steps 47–98). As illustrated in **Figure 1**, the PROCEDURE begins with isolation of GVs from cultures of Ana and Halo, or from *E. coli* expressing a heterologous GV gene cluster from *Bacillus megaterium* (Mega)¹⁵ (note that the Mega GVs were referred to as *E. coli* GVs in our previous work¹¹). This results in three nanostructure populations with distinct properties (**Table 1**). These GVs are then quantified and characterized using pressure-dependent spectroscopy, DLS and TEM. Protocols for modification and functionalization using genetic and chemical approaches are provided as an added option based on the desired end application. Finally, the GVs are imaged *in vitro* and *in vivo* using ultrasound and *in vitro* using HyperCEST MRI.

Advantages and limitations

Among the properties that make GVs attractive as imaging agents are their nanometer size, their fundamental physical stability, their biocompatibility as particles made entirely of protein and their lack of requirement for metals or other inorganic cofactors to produce image contrast. In addition, as the first genetically encodable contrast agents for ultrasound and hyperpolarized xenon MRI, they offer the possibility of being developed as reporter genes¹⁶, creating new opportunities for noninvasive imaging at the molecular and cellular levels.

Among the limitations of GVs in molecular imaging applications are their lower echogenicity per particle as compared with that of conventional microbubble contrast agents for ultrasound and the specialized hyperpolarized Xe approach currently needed to take advantage of their sensitivity as reporters for MRI. In addition, the modular engineering approach presented here is performed *in vitro* on purified GVs, and future efforts will focus on extending this platform to engineering of these nanostructures completely within cells. Furthermore, the use of GVs as dynamic molecular sensors based on conditional clustering requires ascertaining that a given molecular design does not result in nonspecific aggregation, which must be done on a case-by-case basis.

Experimental design

Here, we describe key experimental parameters for each stage of the protocol. One important initial consideration is the choice of GV-producing species. Although this protocol presents methods for producing three different types of GVs—Ana, Halo and Mega—one of these types may be most appropriate for a given application (**Table 1**). For ultrasound, unmodified Halo GVs can be used

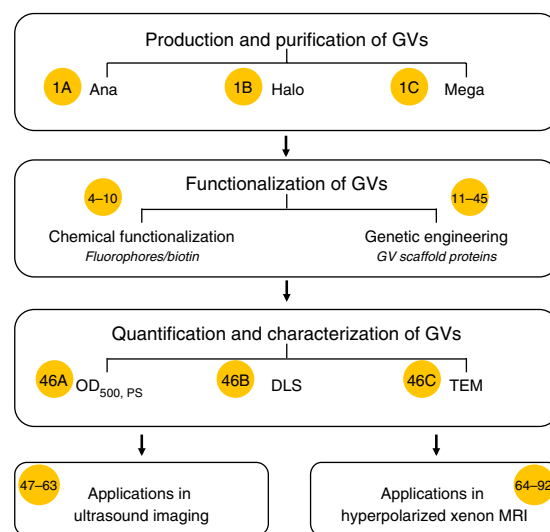


Figure 1 | Flowchart illustrating the experimental design and workflow for gas vesicle (GV) production, quantification, characterization, functionalization and imaging applications (step numbers are given in yellow circles). DLS, dynamic light scattering; OD_{500, PS}, pressure-sensitive optical density; TEM, transmission electron microscopy.

directly in ultrasound imaging to obtain nonlinear signals^{8,10}. Ana GVs are the system of choice if one wishes to genetically tune the properties of GVs for multiplexing, multimodal imaging and targeting applications^{9,14}. Mega GVs produce lower echogenicity under ultrasound as compared with those of Ana and Halo GVs (**Supplementary Fig. 1a–d**), but have a higher critical collapse pressure that may make them useful for multiplexing. Halo GVs produce nonlinear ultrasound contrast immediately after purification, whereas Ana GVs require a chemical treatment. With regard to Xe-MRI, Ana and Mega GVs are more stable under pressure and during the bubbling of hyperpolarized xenon (as described below) as compared with Halo GVs¹¹. All three species have a unique chemical shift in Xe-MRI, allowing multiplexing.

Production and purification of GVs. GVs are obtained from Ana, Halo or heterologously expressing *E. coli*. Ana is a green, filamentous cyanobacterium that naturally inhabits freshwater lakes⁶. Halo is a pink, halophilic species of archaea that grows in saltwater ponds⁷. Ana is cultured in low-salinity medium supplemented with trace metals and buffering agents, whereas Halo is cultured in high-salinity medium for GV production.

Ana and Halo cultures natively produce ample GVs after a few weeks of growth^{8,9,14}. Ana cultures additionally require a controlled gaseous environment and illumination for optimal growth (**Fig. 2a,b**). A freshly inoculated culture of cells may require several rounds (typically 2 or 3) of subculture to become strongly proliferative (**Fig. 2c,d**). The confluent culture of microbes is then transferred to a separatory funnel that is left undisturbed for up to a week to allow the buoyant cells producing GVs to float to the top and separate from media (**Fig. 2e,f**). Buoyant cells are then lysed using hyperosmotic shock for Ana and hypo-osmotic shock for Halo. Subsequently, centrifugally assisted floatation is used to isolate GVs from the cell lysate to yield a concentrated, milky white solution of GVs in the buffer of choice^{8,9,14} (**Fig. 2g–j**).

PROTOCOL

TABLE 1 | Characteristics of different types of GVs.

	Ana GV	Halo GV	Mega GV
Host/origin	<i>Anabaena flos-aquae</i>	<i>Halobacteria salinarum</i>	Heterologous expression of a gene cluster from <i>Bacillus megaterium</i> in <i>E. coli</i> .
Shape	Cylindrical	Spindle	Cylindrical
Resistance to pressure-induced collapse	Medium (can be tuned)	Low	High
Ultrasound contrast	High Nonlinear after engineering	High Nonlinear	Low
Stability in Xe-MRI phantoms	High	Low	High
Ease of genetic modification	High	Low	Not established

Heterologous production of Mega GVs in *E. coli* is accomplished by expression from a plasmid encoding a Mega GV gene, followed by detergent-mediated lysis¹¹.

The procedures leading from inoculation of GV-producing microbes to harvesting and purification are summarized in the Table 2, along with important parameters that affect processing time, yield and quality.

Growth conditions are chosen to facilitate optimal proliferation of each host strain and GV expression. One unusual variable to keep track of is pressure, as GVs collapse irreversibly at hydrostatic pressures of 50–800 kPa, depending on species⁶. For example, the cultures should be grown under mild agitation, as excessive shaking may lead to GV collapse. During centrifugation steps, it is necessary to calculate the hydrostatic pressure generated for a particular *g* force on the liquid column of GVs and to ensure that it is well below the GV critical collapse pressure. Long-term storage of purified GV stocks should preferably be done in screw-top vials, as microcentrifuge tubes with snap-lock caps may cause GV collapse due to pressurization of the sample while opening or closing the tube.

Quantification and characterization of GVs. Purified GVs resuspended in the buffer of choice (e.g., PBS) can be quantified by measuring the OD at 500 nm (OD₅₀₀), as GVs scatter visible light (Step 2). Collapsed GVs (in the same buffer), which do not scatter light, are typically used as the blank control for measurements, yielding a pressure-sensitive OD reading (OD_{500,ps}). It is important to note that clustering of GVs, whether by design or due to functionalization with aggregation-prone moieties, can confound OD₅₀₀ measurements and contribute to errors in calculating concentration from OD₅₀₀. Pressurized absorbance spectroscopy assays GV mechanical strength by measuring OD₅₀₀ under increasing hydrostatic pressure using the device described in the Equipment Setup section (Fig. 3) and the protocol described in Step 46A(i–ix).

DLS is used to estimate the hydrodynamic size of GVs for routine nondestructive characterization and quality control (Step 46B(i–ii)). DLS can be used to assess GV clustering. Care should be taken in the interpretation of DLS readings of GVs because of the spherical assumption of the Einstein–Smoluchowski relation and the nonspherical shape of GVs. Negative-contrast TEM is used for imaging GV size, shape, texture and integrity following production and physical or biochemical treatments (Step 46C(i–viii)). Negative staining with uranyl acetate is used to produce

contrast, and use of a buffer such as HEPES is preferred over use of phosphate buffers that may precipitate with the uranyl acetate. The concentration of the GV solution spotted on the grid directly correlates with the density of GV particles on the grid.

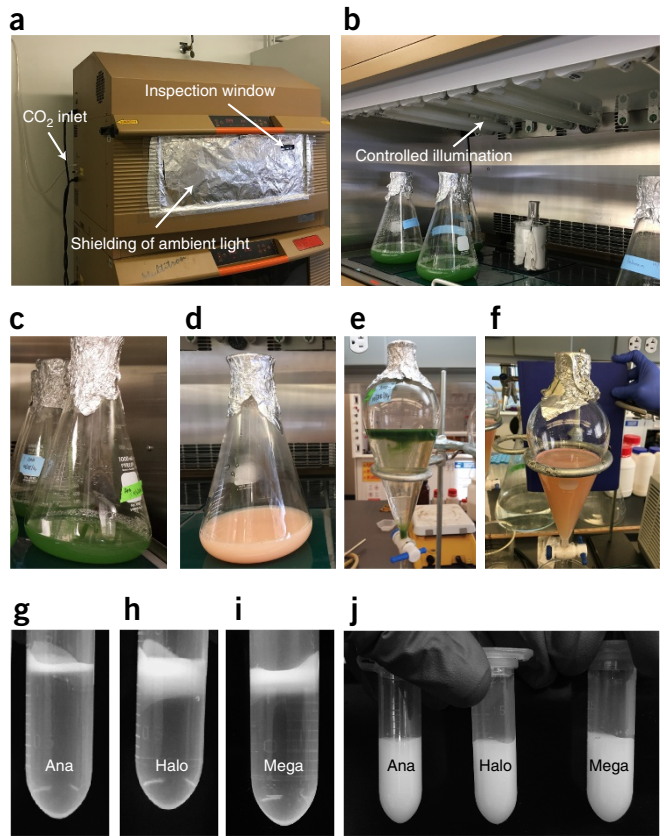


Figure 2 | Equipment setup and expected results for native production and purification of GVs. (a,b) Shaker incubators adapted for the growth of cyanobacteria *Anabaena flos-aquae*, providing controlled illumination, temperature, aeration and CO₂. (c,d) Confluent green (c) and pink (d) cultures of *Anabaena flos-aquae* and *Halobacteria salinarum*, respectively, just before harvesting. (e,f) Effective separation of buoyant cells from spent media in separatory funnels for isolation and harvesting of (e) Ana and (f) Halo cells producing GVs. Purified Ana (g), Halo (h) and Mega (i) GVs as dense milky white layers post centrifugally assisted floatation. (j) Resuspended milky white solutions of purified Ana (left), Halo (middle) and Mega (right) GVs in PBS at OD_{500,ps} ~6 before use in ultrasound imaging experiments.

TABLE 2 | Experimental parameters for GV production, purification and storage.

Procedure	Design parameters
Inoculation of starter culture	Type of culture (suspension vs. solid), amount of inoculum and total volume of culture
Growth of starter culture	Temperature, rotation speed, duration and illumination
Subculturing	Number of flasks, volume of culture and media
Harvesting of GVs	Composition of lysis buffer and duration of lysis, concentration of cells
Purification	Selection of centrifugation speed, type of rotor, tube and syringe needle
Storage	Storage temperature, buffer and type of vial/tube

Modification and functionalization of GVs. GVs can be modified and functionalized through genetic and chemical methods. Genetic engineering of GV mechanical and surface properties is conveniently performed via exchange of the native outer scaffolding protein GvpC (gas vesicle protein C) with recombinant GvpC variants (Steps 11–41). For example, fusing short peptides to the termini of GvpC allows for cell-specific targeting, and truncated GvpC variants enable acoustic multiplexing¹⁴. GvpC tolerates terminal fusions of peptides of up to 100 residues. To covalently attach larger proteins, one can use the modular SpyTag-SpyCatcher protein assembly system¹⁷. GVs reconstituted with SpyTag-GvpC can be reacted with proteins of interest fused to SpyCatcher, forming a covalent bond (Steps 42–45).

Chemical conjugation to GVs makes use of lysine residues on their protein shells and amine-reactive cross-linkers such as sulfo-*N*-hydroxysuccinimide esters (Sulfo-NHS). Chemical moieties, including polymers (e.g., PEG), fluorophores and small molecules (e.g., biotin), can be conjugated using this method (Steps 4–10). Biotinylated GVs can subsequently react with streptavidin or avidinated antibodies⁸. The Sulfo-NHS coupling reaction can be conducted in PBS (pH 7.4) as a one-pot reaction. Depending on the application, the desired extent of labeling can be tuned by varying the molar ratio of Sulfo-NHS to GVs and by changing the incubation time. Either dialysis or buoyancy purification can be used to separate the labeled GVs from excess reactants.

Ultrasound imaging of GVs. For *in vitro* imaging, GVs can be embedded and imaged in multiwell agarose phantoms (Fig. 4a, Steps 47–62). A typical experimental setup for *in vitro* and *in vivo* ultrasound imaging is shown in Figure 4. Experimental design parameters include phantom composition (i.e., percentage of agarose, buffer, background scattering particles), concentration and amount of GV sample loaded. Phantom molds can be made

using 3D printing to obtain a defined well size, shape and spacing. Imaging parameters include plane of imaging (longitudinal versus transverse cross-section of the phantom wells), mode of

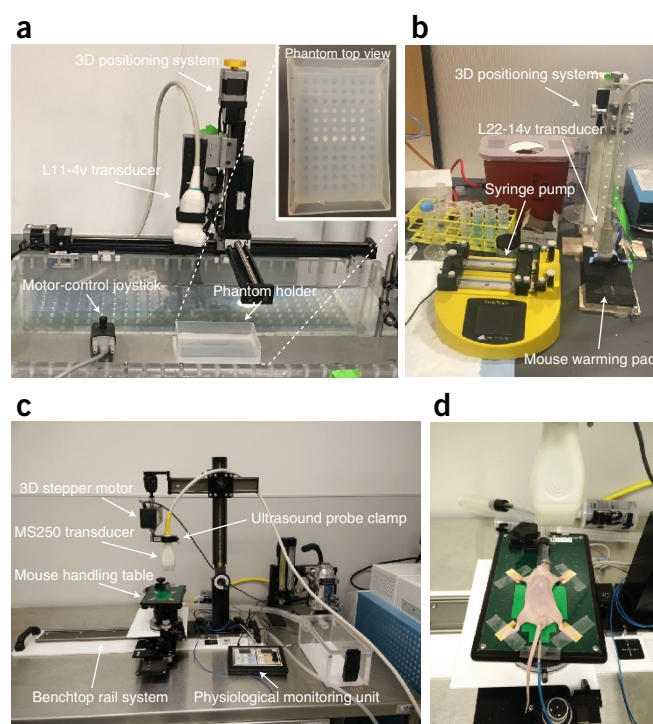


Figure 4 | Ultrasound setup for *in vitro* and *in vivo* imaging. (a) Setup of the *in vitro* imaging system with the Verasonics L11-4v transducer mounted on a 3D translatable clamp (computer-controlled) and a fixed imaging phantom holder. Inset shows an agarose phantom loaded with GV and polystyrene samples before imaging. (b) *In vivo* imaging setup with the Verasonics L22-14v transducer, mounted on a 3D positioning system (manual), an animal-mounting platform with a heating pad and a syringe pump for controlled sample injections. (c) Setup of the Vevo imaging station includes the ultrasound probe clamp attached to the 3D stepper, the mouse handling table and the physiological monitoring unit. All components except the physiological monitoring unit are attached to the benchtop rail system. (d) Animal positioning, with the mouse being laid down in a supine position with the nose and mouth in the nose cone and each paw extending outward onto the electrodes and secured with surgical tape. A rectal probe is used to monitor the core temperature of the animal. All procedures involving animal use must be performed in accordance with institutional guidelines and regulations and approved by relevant animal care and use committees. All animal experiments reported in this article received approval from the institutional animal care and use committees of the California Institute of Technology or the Sunnybrook Research Institute.

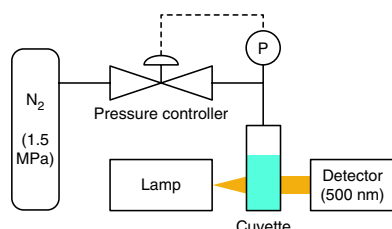


Figure 3 | Collapsometry setup. Illustration of the collapsometry setup used for determining the critical collapse pressure of GVs. Image adapted with permission from ref. 10, Elsevier.

ultrasound imaging (conventional or nonlinear imaging), transducer frequency range, transmit waveform characteristics such as pulse envelope shape, number of cycles, amplitude and frequency. Importantly, the amplitude used for imaging must be below the GV's acoustic critical collapse pressure. PBS is typically used as a negative control for ultrasound contrast and 5- μm polystyrene beads that scatter linearly at medical ultrasound frequencies are used as a reference sample. All GV samples and controls are mixed with melted agarose solution before loading. Solidification of the agarose after loading into the phantom wells ensures that samples are uniformly distributed throughout the well and that GVs remain suspended in the agarose matrix without floating during imaging. Typically, a final OD₅₀₀ value of 2.25 for Ana GVs and 0.4% (wt/vol) for polystyrene beads is used to match echogenicity. For Halo GVs, a final OD value of 0.5 gives good signal without attenuation, with 0.83% (wt/vol) polystyrene beads to match echogenicity. *In situ* collapse of GVs using pulses with amplitudes above the GVs' acoustic critical collapse pressures cause GV ultrasound signals to disappear, allowing confirmation of GV-based signals and background subtraction. In addition, differential collapse at multiple pressures and spectral unmixing allow multiplexed imaging of mechanically distinct GV classes¹⁴. Quantification of signals in ultrasound images is typically performed using MATLAB or ImageJ software.

The *in vivo* ultrasound imaging protocol has been used to look at GV passage through the inferior vena cava (IVC) and subsequent induced contrast enhancement in the liver after i.v.

injection^{8,14} (Step 63A–B), but it can be adapted for other organs or tissues. Depending on the tissue region of interest, some ultrasound imaging parameters, such as image gain and field of view, may need to be adjusted accordingly. As in *in vitro* imaging, it is critical that the transmit power be kept at a value that provides sufficient signal without collapsing the GVs. Further adjustments can be made to GV solution concentration and volume, according to experimental needs. Functionalized GVs can also be imaged using this protocol, as long as appropriate controls are used. When planning an imaging experiment using functionalized GVs, we suggest using native GVs from the same batch as a control. It is possible to administer multiple injections of GVs in the same mouse, as long as the total injection volume does not exceed the limit stated in institutional guidelines. If required, GV solutions can be tested for bacterial endotoxins using quantitative, chromogenic endpoint LAL assays, such as the QCL-1000 Assay (Lonza). Endotoxins can potentially be removed using commercially available affinity resins such as the ToxinEraser Endotoxin Removal Resin (GenScript). Once injected, ensure that GVs have cleared completely and the tissue contrast signal is back to baseline before a second bolus injection.

¹²⁹Xe MRI imaging of GVs. Hyperpolarized ¹²⁹Xe-MRI, an active area of molecular imaging research, is currently mainly performed *in vitro* with pure contrast agents or labeled cells^{18,19} while *in vivo* techniques are being developed^{12,13}. This protocol describes the imaging of GVs using hyperpolarized ¹²⁹Xe MRI

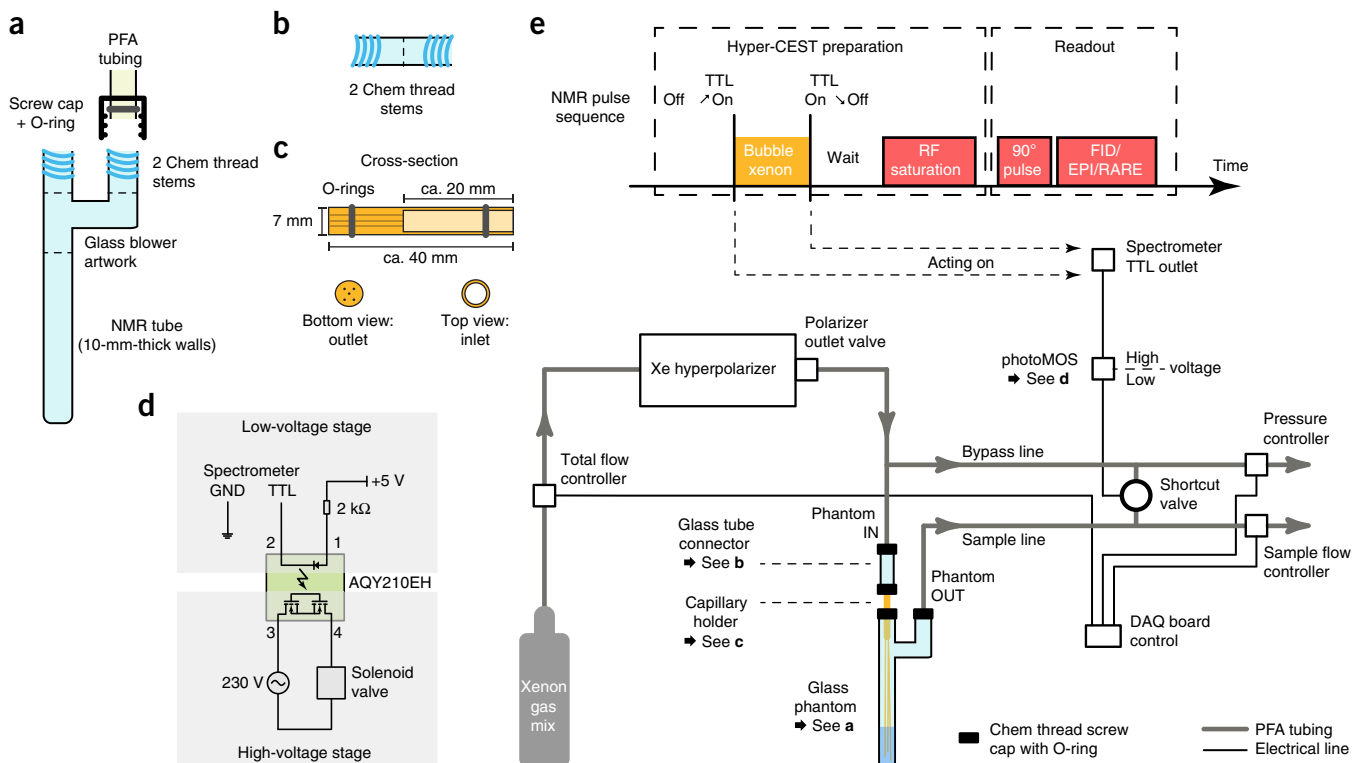


Figure 5 | Required elements for assembly of the gas-delivery manifold for ¹²⁹Xe-MRI. (a) Main glass phantom made from an NMR tube (10 mm, thick walls) and chem thread connectors; PFA tubings are connected with a chem thread screw cap and an O-ring. (b) Glass tube connector (this is also used for Step 67, but mainly for connecting the PFA tubing with the capillary holder). (c) Capillary holder. (d) PhotoMOS circuit for controlling the solenoid valve. (e) Schematics of the communication between the NMR pulse sequence and the gas-delivery manifold, with corresponding tube and electrical connections. DAQ, data acquisition.

in vitro, as previously done with purified GVs, GV-expressing cells and GV-labeled cells¹¹. This involves preparing the hyperpolarized gas, introducing it into a liquid sample of GVs and applying a HyperCEST pulse sequence¹¹ (Steps 64–92). This requires some specialized equipment, some of which has been described earlier in the context of xenon NMR and MRI applications^{11,20,21}. The working steps and the equipment setup in particular require people with skills in NMR/MRI and some experience in setting up custom-made electronic devices. The hyperpolarization system is available as a commercial unit (see Equipment section and

Fig. 5) or can be built according to previous publications²². In either case, the system should be capable of delivering hyperpolarized ¹²⁹Xe under very stable, continuous flow conditions^{20,21}.

To achieve good Hyper-CEST performance with a particular GV type, it is necessary to optimize the saturation power and duration to match the exchange conditions for Xe between GVs and the surrounding media^{23,24}. In addition, attention must be paid to GV integrity during the bubbling or inflow of xenon into the phantom, as both hydrostatic pressure and bubbles can cause irreversible GV collapse.

MATERIALS

REAGENTS

- *Anabaena flos-aquae*/*Aphanizomenon flos-aquae*/*Dolichospermum flos-aquae* (Culture Collection of Algae and Protozoa (CCAP), Scottish Marine Institute, cat. no. CCAP 1403/13F)
- *Halobacterium* sp. NRC-1 as brine inclusions (Carolina Biological Supply, cat. no. 154777) or *Halobacterium* sp. plate (Carolina Biological Supply, cat. no. 154801)
- Rosetta 2(DE3) pLysS competent *E. coli* cells (EMD Millipore, cat. no. 71401-3)
- BL21 (DE3) competent *E. coli* cells (NEB, cat. no. C25271)
- Cyanobacteria BG-11 Freshwater solution, 50× (Sigma-Aldrich, cat. no. C3061)
- PBS without calcium and magnesium (Corning, cat. no. 21-040)
- Sodium nitrate, high purity grade (NaNO₃; Amresco, cat. no. 0598)
- Potassium phosphate monobasic (KH₂PO₄; Sigma-Aldrich, cat. no. P5655)
- Magnesium chloride hexahydrate (MgCl₂·6H₂O; Sigma-Aldrich, cat. no. M9272)
- Trisodium citrate dihydrate (Sigma-Aldrich, cat. no. S1804)
- Potassium chloride (KCl; Sigma-Aldrich, cat. no. 746436)
- Casein hydrolysate (Sigma-Aldrich, cat. no. 22090)
- Yeast extract (BD, cat. no. 212750)
- Magnesium sulfate heptahydrate (MgSO₄·7H₂O; Sigma-Aldrich, cat. no. 230391)
- Calcium chloride dihydrate (CaCl₂·2H₂O; BDH Chemicals, cat. no. BDH9224)
- Sodium metasilicate nonahydrate (Na₂SiO₃·9H₂O; Sigma-Aldrich, cat. no. S4392)
- Sodium carbonate (Na₂CO₃; Sigma-Aldrich, cat. no. 71345)
- Sodium bicarbonate (NaHCO₃; Sigma-Aldrich, cat. no. S5761)
- Citric acid, 1 M (BDH Chemicals, cat. no. BDH7397)
- EDTA, disodium salt dihydrate (Alfa Aesar, cat. no. A15161)
- Ferric ammonium citrate (Amresco, cat. no. 0846)
- Solulys bacterial protein extraction reagent, in Tris-HCl buffer, pH 7.4 (Genlantis, cat. no. L200500)
- D-Sorbitol (Sigma-Aldrich, cat. no. S1876)
- ddH₂O (tap supply)
- Anhydrous DMSO (Life Technologies, cat. no. D12345)
- N-hydroxy-succinimidyl-ester moiety such as Alexa-488-NHS (Thermo Fisher, cat. no. A20000)
- Tris-HCl, 1M, pH 8.0 (Teknova, cat. no. T1080)
- HEPES (Sigma-Aldrich, cat. no. H3375)
- Terrific broth (Sigma-Aldrich, cat. no. T0918)
- Isopropyl β-D-1-thiogalactopyranoside (IPTG; Anatrace, cat. no. I1003)
- Kanamycin sulfate (Sigma-Aldrich, cat. no. K1377)
- 2% (wt/vol) Uranyl acetate solution (EMS, cat. no. 22400-2)
- Urea, high-purity grade (VWR, cat. no. 0568)
- Sodium hydroxide (NaOH; Sigma-Aldrich, cat. no. 221465)
- Sodium chloride (NaCl; Sigma-Aldrich, cat. no. 793566)
- Sodium phosphate monobasic, monohydrate (NaH₂PO₄·H₂O; Amresco, cat. no. 0823)
- 10× Tris/Glycine/SDS buffer (Bio-Rad, cat. no. 1610732)
- Precision Plus Protein Kaleidoscope Standards (Bio-Rad, cat. no. 1610375)
- Laemmli sample buffer (Bio-Rad, cat. no. 1610737)
- PowerPac Basic Power Supply (Bio-Rad, cat. no. 1645050)
- Mini-Protein Tetra Cell for Mini Precast Gels (Bio-Rad, cat. no. 1658004)
- Any kD Mini-Protein TGX Precast Protein gels (Bio-Rad, cat. no. 4569034)

- SimplyBlue SafeStain (Invitrogen, cat. no. LC6060)
- Bradford reagent (Sigma-Aldrich, cat. no. B6916-500ML)
- Pierce 660nm protein assay reagent (Thermo Fisher Scientific, cat. no. 22660)
- BSA (Thermo Fisher Scientific, cat. no. 15561020)
- Ni-NTA Superflow Resin (Qiagen, cat. no. 30450)
- Poly-Prep chromatography columns (Bio-Rad, cat. no. 7311550)
- Imidazole (Sigma-Aldrich, cat. no. 792527)
- 2-Mercaptoethanol (Bio-Rad, cat. no. 1610710)
- DNaseI (NEB, cat. no. M0303S)
- Lysozyme, from chicken egg white (Sigma-Aldrich, cat. no. L6876)
- GvpC_{WT}, ΔN&C-GvpC and GvpC-SpyTag genes in pET28a plasmid (Addgene, ID nos. 85732, 85736 and 85737, respectively)
- SpyCatcher-mNeonGreen in pET26b plasmid (<https://www.addgene.org/vector-database/9135>)
- pST39 plasmid containing pNL29 Mega GV gene cluster (Addgene, ID no. 91696)
- SOC outgrowth medium (NEB, cat. no. B9020S)
- Luria Broth (LB) base (Miller's LB Broth Base), powder (Invitrogen; Life Technologies, cat. no. 12795-084)
- Ampicillin sodium salt (Sigma-Aldrich, cat. no. A0166-25G)
- Chloramphenicol (Sigma-Aldrich, cat. no. C0378)
- D-(+)-Glucose (Sigma-Aldrich, cat. no. G8270)
- Glycerol (Sigma-Aldrich, cat. no. G5516)
- PBS, 20× solution (Teknova, cat. no. P1225)
- SCID Hairless Outbred (SHO) or SCID Hairless Congenic (SHC) mice, female, 4–6 weeks of age (Charles River, cat. no. 474 or 488) **! CAUTION** All procedures involving animal use must be performed in accordance with institutional guidelines and regulations and approved by the relevant animal care and use committees. All animal experiments reported in this article received approval from the institutional animal care and use committees of the California Institute of Technology or the Sunnybrook Research Institute.
- Sterile PBS, pH 7.4 (Life Technologies, cat. no. 10010-023)
- Isoflurane, USP (Fresenius Kabi, cat. no. M60303)
- Medical air, USP cylinder (Praxair, DIN 02014483) **! CAUTION** Gaseous anesthesia must be scavenged properly using a charcoal canister or house vacuum to minimize potential exposure to humans.
- Charcoal canister (F-Air; Baintree, cat. no. F-AIR EA)
- Eye ointment (Lacri-Lube, Refresh) or lubricating gel (Aquagel; Parker Laboratories, cat. no. 57-05)
- 0.9% (wt/vol) Sodium chloride injection, (Baxter, cat. no. JB1322 or VWR, cat. no. 47729-565)
- Ultrasound gel (WavelengthCL; National Therapy Products, cat. no. NTPC500X) or Aquasonic 100 ultrasound transmission gel (Parker Laboratories; VWR, cat. no. 68200-712)
- Electrode gel (Signa gel, Parker Laboratories, cat. no. 15-25)
- Nair lotion (Church & Dwight)
- Vetbond tissue adhesive glue (Fisher Scientific, cat. no. NC9259532)
- Sphero polystyrene particles, 4.5–4.9 μm (Spherotech, cat. no. PP-45-10)
- Agarose (Bio-Rad, cat. no. 161-3102)
- Xenon gas mixture, 2–5% (vol/vol) Xe (natural abundance with 26.4% (vol/vol) ¹²⁹Xe is sufficient), 10% (vol/vol) N₂, He balance, grade 5.0 (Praxair) **▲ CRITICAL** The gas tank should have a pressure regulator for adjustments ~2 bar absolute pressure. The gas amount to be used

depends on the flow rate through the system. A typical value would be 0.25 liters/min—i.e., 15 liters/h.

EQUIPMENT

- Shaker incubator with cooling, CO₂ control and illumination (Infors HT, model no. Multitron II; exact setup used is shown in **Fig. 2a–b**)
- Benchtop centrifuge with swinging-bucket rotor (Beckman Coulter, model no. Allegra X-15R)
- Benchtop microcentrifuge (Eppendorf, model no. 5424R)
- Barnstead GenPure xCAD Plus UV/UF Water Purification System (Thermo Scientific, cat. no. 50136146)
- Pyrex 500-ml and 1,000-ml Erlenmeyer flasks (Corning, cat. no. 4980)
- Pyrex 500-ml separatory funnel (cat. no. 6402-500)
- Pyrex culture tubes (Corning, cat. no. 9820)
- Pyrex 250-ml, 500-ml and 1-liter media storage bottles (Corning, cat. nos. 1395-250, 1395-500 and 1395-1L)
- 4-liter beaker (Nalgene; Thermo Scientific, cat. no. 1201-4000)
- 15- and 50-ml centrifuge/falcon tubes
- 1.5-, 1.7- and 2-ml microcentrifuge tubes
- Metal ring stand for separatory funnel
- Vacuum-driven disposable bottle-top filter (Durapore, 0.22 µm; Millipore, cat. no. SCGVT05RE)
- Adjustable-volume pipettes and pipette tips (100–1,000, 20–200, 2–20 and 0.5–2.5 µl; Eppendorf Research Plus)
- Serological pipettes (1, 5, 10 and 25 ml, VWR)
- Syringes, with Luer-Lok tips, 1, 3, 10 and 30 ml (VWR, cat. nos. 309602, 309657, 309604 and 309650)
- 1-inch, 18-gauge and 1/1.5-inch 21-gauge bulk blunt needles (SAI Infusion Technologies, cat. nos. B18-100 Bulk and B21-100/B21-150 Bulk)
- Gel-loading pipette tips (Corning, cat. no. 4884)
- Regenerated cellulose dialysis tubing, 6–8 kDa cutoff (Spectra/Por, cat. no. 132665)
- Formvar/Carbon 200 mesh Cu TEM grids (Ted Pella, cat. no. 01801)
- Ultracentrifuge (Beckman-Coulter, model no. Avanti J-30L) with rotor JLA-16.250 (part no. 363930)
- Polypropylene bottle with cap assembly (Beckman-Coulter, cat. no. 357001) with required adaptor (Beckman-Coulter, part no. 356997)
- PD10 desalting columns (GE Healthcare Life Sciences, cat. no. GE17-0851-01)
- 1-cm-path-length quartz cuvette (Hellma Analytics, cat. no. 176.700-QS)
- Pressure controller (Alicat Scientific, cat. no. 73907)
- UV-visible spectrophotometer (Ocean Optics, STS series microspectrometer, 350–800 nm, 50-µm slit, 400 core/1-cm SMA input, cat. no. STS-VIS-L-50-400-SMA) equipped with a 400-µm fiber in the VIS/NIR range (2 m; Ocean Optics, cat. no. P400-2-VIS-NIR) and a Krypton light source (2,000-h bulb, ecoVis) and accompanying Oceanview spectroscopy software
- Compressed nitrogen cylinder with pressure regulator (Air Liquide, cat. no. UN 1066) **! CAUTION** Proper safety techniques should be followed when opening and handling the compressed gas cylinder. Make sure that the cylinder is secured to a wall and properly connected to the pressure regulator before opening.
- ZetaPALS DLS instrument (Brookhaven Instruments)
- Disposable cuvette (Eppendorf, cat. no. Z605050)
- Plate-reader spectrophotometer (Molecular Devices, model no. SpectraMax M5)
- Gel imaging system (ChemiDoc MP; Bio-Rad, model no. 1708280)
- Water bath (e.g., Fisher Scientific, cat. no. 15-462-6Q)
- Disposable culture tube, 16 × 100 mm (VWR, cat. no. 47729-576)
- Dialysis clips, weighted/unweighted closures, 55-mm sealing width (Spectrum Labs, cat. no. 132745/132737)
- Transmission electron microscope (Thermo Fisher FEI, model no. Tecnai 120 keV T12 LaB6) equipped with a Gatan Ultrascan 2k × 2k charge-coupled device (CCD) and Gatan Digital Micrograph data collection software suite (<http://www.gatan.com/installation-instructions>)
- Glow discharging system (Emitech, model no. K100X)
- Pelco reverse, anti-capillary tweezers (Ted Pella, cat. no. 5378-NM)
- Glass slides (microslides; VWR, cat. no. 48382-171)
- Filter paper (Whatman, cat. no. 1001-090)
- Pipet-Aid (Drummond Scientific, cat. no. DRCT011)
- Magnetic stirring bars (VWR, cat. no. 58948-193)
- Rotatory shaker (Rotamix; Appropriate Technical Resources, model no. RKVS)
- Microvolume spectrophotometer (NanoDrop 2000C series;

- Thermo Scientific, cat. no. ND-2000C)
- Tail-vein catheter (SAI Infusion Technologies, cat. no. BF-27-01)
- Syringe with tip cap, 12 ml (Luer Lock Kendall Monoject; Covidien, cat. no. 1181200777)
- Needle, 27-gauge × 1/2 (PrecisionGlide; BD, cat. no. 305109)
- Tuberculin syringe, 12 cc (BD, cat. no. 305620)
- Surgical tape (Transpore; 3 M, cat. no. 1527-1)
- Alcohol wipes (Webcol; Covidien, cat. no. 63001) or (BD, cat. no. BD326895)
- Cotton swabs
- Syringe pump (New Era Pump System, cat. no. NE-1000 or Kent Scientific Corporation, model. GenieTouch; typical setup shown in **Fig. 4b**)
- Heat lamp (Home Hardware)
- Ultrasound imaging system (Fujifilm VisualSonics, model no. Vevo2100 or Verasonics, model no. Vantage 256; typical setup shown in **Fig. 4a–d**)
- Ultrasound transducer (Verasonics, model no. L11-4v or L22-14v, or Fujifilm VisualSonics, model no. MS250)
- Acoustic absorber sheet, 300 × 300 mm (Precision Acoustics, cat. no. F28-SMALL)
- Multiwell phantom mold (3D-printed)
- Plastic reservoir for phantom preparation (flat-bottom polypropylene robotic reservoirs, Nalgene, cat. no. 1200-1300)
- Computer-controlled translation stage (X-Slide, Velmex).
- MATLAB (MathWorks, https://www.mathworks.com/programs/trials/trial_request.html?ref=ggl&ss_eid=ppc_5852767762&q=matlab%20download)
- MATLAB scripts (see **Supplementary Data**)
- Animal heat pad (Braintree Scientific, cat. no. HP 1M)
- Temperature controller (Braintree Scientific, cat. no. TCAT 2DF)
- Rectal probe for mice (Braintree Scientific, cat. no. RET 3)
- Vevo Imaging Station (Fujifilm VisualSonics)
- PE10 plastic tubing (Becton Dickinson, cat. no. 427400)
- 30-gauge needles (Becton Dickinson, cat. no. 305106)
- Rodent anesthesia machine (Patterson Veterinary, cat. no. 07-870-3592 or VetEquip, cat. no. 901806)
- Weighing balance (sensitive in the gram and milligram range)
- MR imaging system (Bruker); preferably a vertical NMR spectrometer at a field strength of 7 T or higher with micro-imaging capability for ¹H and ¹²⁹Xe; a variable temperature unit (VTU) is desirable to keep the sample temperature stable either at room temperature (18–25 °C) or at physiological body temperature **▲ CRITICAL** The operating console must have at least one available trigger signal output (TTL) to be connected to a photoMOS relay for regulating the gas flow.
- Dual-tuned ¹H/¹²⁹Xe NMR coil with 10-mm inner diameter (Bruker)
- Nonselective NMR pulse sequence for acquiring a direct Xe NMR spectrum after fresh Xe delivery. This is implemented by using a simple (pulse-acquire) sequence; the sequence starts with two trigger signals to communicate with the gas-delivery manifold and then a block excitation pulse of adjustable amplitude, followed by a free induction decay (FID) acquisition
- Multi-echo acquisition MRI pulse sequence (echo planar imaging (EPI) or rapid acquisition with relaxation enhancement (RARE), Bruker) with activatable magnetization transfer module (alternative: fat suppression or saturation transfer module) and two triggers
- Hyperpolarization system for producing spin-hyperpolarized ¹²⁹Xe (commercial systems are available through <http://xemed.com> and <http://www.polarean.com> or as an open-source design) **! CAUTION** The hyperpolarization system contains ~1 g of a highly reactive alkali metal such as Rb. Make sure to become familiar with the procedures for handling the system as provided by the manufacturer **▲ CRITICAL** The system must be capable of producing hyperpolarized Xe in continuous flow mode for repeated re-delivery into the sample solution.
- Gas flow controller with a maximum flow rate of 100 standard ml/min (SMLM; Omega Engineering, model no. FMA 5408) and second one of 500 SMLM flow (Omega Engineering, model FMA5412) **▲ CRITICAL** Such gas-flow controllers are typically calibrated for N₂ gas; the correction factors for systems based on heat flux measurements applied to the Xe gas mix are as follows: 1.44 for the Xe fraction and 1.454 for the He fraction. Hence, the correction factor for a 5% (vol/vol) Xe mix is 0.05 × 1.44 + 0.1 × 1 + 0.85 × 1.454 = 1.408.
- Pressure controller for up to 6 bar and 1 standard liter per minute flow (Bronkhorst, model no. P-702CV-6K0A-AAD-22-V)
- Direct solenoid-actuated poppet ‘shortcut’ valve (Norgren, model no. 9502310)

- PhotoMOS relay (Panasonic, model no. AQY210EH)
- PC-operated control interface for the hyperpolarization system and the gas flow controllers based on a data acquisition board (Omega Engineering, model no. Daq/board 3000) and a graphic user interface (e.g., National Instruments, model DASYLab or LabView)
- Glass phantom (made from thick-wall NMR tube of 10-mm outer diameter; Wilmad) with inlet and outlet adaptor with size 7 chem threads (ChemGlass Life Sciences, cat. no. CG-350-01); see **Figure 5a**
- 5-mm standard NMR tube to be inserted into the glass phantom to create a second compartment inside the 10-mm NMR tube
- 1/4-inch PFA flexible tubing (Swagelok) **▲ CRITICAL** Ensure that the PFA tubing directing the Xe flow from the hyperpolarizer into the sample does not go through any zero-field crossings in the fringe field of the superconducting magnet because this might cause a loss of hyperpolarization.
- Fused silica glass capillaries for Xe dispersion (e.g., five pieces of ~15-cm length each and 350- μ m outer diameter, e.g., Polymicro, material no. 106815-0030)
- Glass connecting tube with size 7 chem threads on either end (**Fig. 5b**)
- Home-made capillary holder; ~40 mm long, having a 20-mm-deep, 5-mm-diameter bore on the inlet side and five clear bores matching the outer diameter of the capillaries (as in **Fig. 5c**; Teflon or Delrin material) on the outlet side
- pH meter
- Magnetic stir plate (Thermo Scientific, cat. no. S194925)
- Autoclave
- Bunsen burner
- Spark lighter

REAGENT SETUP

G625 growth medium for Ana (4 liter) G625 growth medium is 5.84 mM NaNO_3 , 224 μM KH_2PO_4 , 304 μM $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 208 μM $\text{Na}_2\text{SiO}_3 \cdot 9\text{H}_2\text{O}$, 189 μM Na_2CO_3 , 10 mM NaHCO_3 , 245 μM CaCl_2 , 31 μM citric acid and 3 μM EDTA. Separately, add 24 mg of ferric ammonium citrate to 1 ml of ultrapure water (in a 1.5-ml microcentrifuge tube). Heat the mixture in a 42 °C water bath for 10–15 min to dissolve the ferric ammonium citrate before adding to the medium in a Nalgene 4-liter beaker. Adjust the pH to 8.0 with HCl. Filter-sterilize using a 0.22- μm bottle-top filter. G625 growth medium prepared using this method is stable for at least 6 months if stored in a cool, dry place at room temperature under sterile conditions. Because the iron sediments, make sure that the bottle is swirled each time before use.

Carolina growth medium for Halo (1 liter) Carolina growth medium is 4.278 M NaCl, 81 mM $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 10 mM trisodium citrate dihydrate, 27 mM KCl, 5g/liter casein hydrolysate and 3g/liter yeast extract. Adjust the pH to 7.2 with NaOH and autoclave at 121 °C and 18 psi for 30 min. Autoclaved medium can be stored for at least 6 months under sterile conditions at room temperature.

TMC lysis buffer for Halo GVs TMC lysis buffer is 10 mM Tris-HCl, 2.5 mM MgCl_2 and 2mM CaCl_2 , pH 7.5. The buffer should be filtered (0.22- μm filter) and can be stored at room temperature indefinitely.

Sorbitol lysis buffer for Ana GVs Sorbitol lysis buffer is 1 M D-sorbitol in dH_2O . The buffer should be filtered (0.22- μm filter) and can be stored at room temperature indefinitely.

Terrific broth Add 47.6 g of terrific broth (TB) powder and 8 ml of glycerol to 1 liter of dH_2O . Autoclave at 121 °C for 15 min. Autoclaved medium can be stored for at least 6 months under sterile conditions at room temperature.

LB medium Add 25 g of LB powder to 1 liter of dH_2O . Autoclave at 121 °C for 15 min. Autoclaved medium can be stored for at least 6 months under sterile conditions at room temperature.

Kanamycin stock solution Kanamycin stock solution is 50 mg/ml kanamycin in dH_2O . The solution is at 1,000 $\times$ concentration and should be filter-sterilized after preparation; it can be stored for up to 6 months at –20 °C.

Ampicillin stock solution Ampicillin stock solution is 100 mg/ml ampicillin in dH_2O . The solution is at 1,000 $\times$ concentration and should be filter-sterilized after preparation; it can be stored for up to 3 months at –20 °C.

Chloramphenicol stock solution Chloramphenicol stock solution is 25 mg/ml chloramphenicol in ethanol. The solution is at 1,000 $\times$ concentration and should be filter-sterilized after preparation; it can be stored for up to 6 months at –20 °C.

IPTG inducer stock solution IPTG inducer stock solution is 1 M IPTG in dH_2O . The solution is at 1,000 $\times$ concentration and should be filter-sterilized after preparation and can be stored at –20 °C in the dark for up to 1 year because it is light-sensitive.

GV stripping buffer (Round 1) GV stripping buffer is 10 M urea and 100 mM Tris-HCl, pH 8.0–8.5. This solution should be freshly made before use.

GV stripping buffer (Round 2) GV stripping buffer is 6 M urea and 60 mM Tris-HCl, pH 8.0–8.5. This solution should be freshly made before use.

TGS buffer TGS buffer is a 1:9 (vol/vol) dilution of 10 $\times$ TGS stock in ddH_2O . Store 10 $\times$ TGS stock (as purchased) at room temperature and dilute to 1 $\times$ just before use. Do not use beyond the printed expiry date.

Inclusion body solubilization buffer Inclusion body solubilization buffer is 6 M urea, 20 mM Tris-HCl and 500 mM NaCl, pH 8.0. The buffer should be filter-sterilized and stored at 4 °C. It is not advisable to use this buffer more than a week old for protein purification.

Inclusion body wash buffer Inclusion body wash buffer is 6 M urea, 20 mM Tris-HCl, 500 mM NaCl and 20 mM imidazole, pH 8.0. The buffer should be filter-sterilized and stored at 4 °C. It is not advisable to use this buffer more than a week old for protein purification.

Inclusion body elution buffer Inclusion body elution buffer is 6 M urea, 20 mM Tris-HCl, 500 mM NaCl and 250 mM imidazole, pH 8.0. The buffer should be filter-sterilized and stored at 4 °C. It is not advisable to use this buffer more than a week old for protein purification.

Soluble protein wash buffer Soluble protein wash buffer is 50 mM NaH_2PO_4 , 300 mM NaCl, 20 mM imidazole and 1 mM 2-mercaptoethanol, pH 8.0. The buffer should be filter-sterilized and can be stored at 4 °C for up to 6 months.

Soluble protein elution buffer Soluble protein elution buffer is 50 mM NaH_2PO_4 , 300 mM NaCl and 250 mM imidazole, pH 8.0. The buffer should be filter-sterilized and can be stored at 4 °C for up to 6 months.

Ultrasound gel preparation for imaging experiments Take a 30-ml Luer-Lok syringe, remove the piston and place the syringe with the tip facing down in a 50-ml Falcon tube. Fill the syringe with ultrasound gel before loading the Falcon tube into a table-top centrifuge. Spin at 2,500g for 30 min to ensure that there are no air bubbles in the gel, and keep it at room temperature. It is advisable to prepare the gel on the same day of the experiment, as longer storage may lead to drying if the syringe is not sealed tightly.

EQUIPMENT SETUP

Cleaning and sterilization of glassware Clean and sterilize all glassware by autoclaving, and fully dry all glassware before use.

Setup of the apparatus for pressurized absorbance spectroscopy The setup used to conduct pressurized absorbance spectroscopy is illustrated in **Figure 3** and includes the following components: (i) computer equipped with MATLAB to run the collapse pressure scripts, (ii) compressed N_2 tank with control valves and a regulator, (iii) pressure controller, (iv) flow-through quartz cuvette and (v) UV-Vis spectrometer equipped with a light source and a cuvette holder.

Couple the pressure controller to the gas tank via a stainless-steel gas connector, to the computer via a USB port and to the flow-through cuvette via plastic cannulae. Connect the ecoVis Krypton light source of the UV-visible spectrometer to a power supply, allowing light to pass through the sample in the quartz cuvette. The transmitted light then passes through an optical fiber connector into the OceanOptics STS microspectrometer. The spectrometer is connected to the computer and controlled by a MATLAB script, which also interfaces with the pressure controller (MATLAB scripts can be found in the **Supplementary Data**). An equilibration time of 7 s at each pressure step and a final collapse pressure of 1,400 kPa (for blanking) is hard-coded into the script and kept constant for all measurements. Before acquiring sample measurements, the system is calibrated by recording the spectra for full transmission (using the cuvette loaded with PBS or a buffer of choice) and zero transmission (by turning the cuvette 90° so that no light passes through). **! CAUTION** Ensure that the compressed nitrogen tank is transported and stored safely. Cylinders should be secured and stored upright in a cool, dry area protected from combustible materials. The gas valve seal must always be in place. When not in use, keep all the cylinder and regulator valves shut and the pressure controller and spectrometer powered off.

Setting up the Verasonics imaging station and imaging parameters The Verasonics imaging system and setup for *in vitro* and *in vivo* experiments are shown in **Figure 4a,b**; the system comprises the following components: (i) computer, (ii) ultrasound probe, (iii) translatable probe clamp equipped with three independent motors that move in the x, y and z directions with micrometer-level precision, (iv) fixed phantom holder and (v) Verasonics scripts running on MATLAB software. Adapt the setup for *in vivo* imaging by

replacing the phantom holder with a mouse-mounting platform equipped with a heating pad, anesthesia equipment and nose cone for the animal and a manually translatable probe clamp. **▲ CRITICAL** Make sure that the probe is well coupled to the sample and that the transducer array of the probe is immersed in liquid or covered with ultrasound gel before transmitting ultrasound.

GVs can be imaged using the Ray-lines Verasonics example script. Imaging is typically performed at ~18 MHz using the L22-14v transducer with an F number of 2 or 3, focal depth of 8 mm (matching the natural focus of the transducer) and 6 cycle pulses. For these parameters, nondestructive imaging of GV's can be performed at voltages 3 V. The onset of GV collapse usually takes place at ~5 V. For efficient GV destruction, a 25 V pulse is applied for a few seconds.

GVs are imaged at lower frequencies (~6.25 MHz) using the L11-4v transducer and the corresponding Ray-lines script. The F number is set to 2, transmit focus to 20 mm and transmit voltage to 1.6 V. Images are saved from the Verasonics graphical user interface.

Setup of the Vevo imaging station and imaging parameters The imaging station setup is depicted in **Figure 4c,d**, and encompasses the following components: (i) benchtop rail system; (ii) 3D stepper motor; (iii) ultrasound probe clamp; (iv) mouse-handling table; and (v) physiological monitoring unit. Make sure that the 3D motor is connected to the Vevo2100 ultrasound system to allow for 3D image acquisition. Also connect the physiological monitoring unit to the Vevo2100 system to allow on-screen tracking of physiological parameters such as heart rate, respiratory rate and body temperature.

Halo GV's are imaged under the following imaging settings using the Vevo2100 system: (i) frequency—21 MHz and 18 MHz for B-mode and contrast mode, respectively; (ii) transmit power—2%; and (iii) gain—16dB and 32dB for B-mode and contrast mode, respectively. The imaging field of view can be adjusted accordingly to fit the tissue region of interest, by increasing or decreasing the image width and depth. **▲ CRITICAL** It is critical to keep the transmit power on the Vevo2100 system to 2%, to allow for stable imaging of Halo gas vesicles. Higher transmit power will collapse exposed GV's.

Setting up the syringe pump GV injections are administered as a bolus using a syringe pump (shown in **Fig. 4b**). The following settings are used for each injection: (i) syringe diameter—3.55 mm (inner diameter of a BD 12-cc tuberculin syringe); (ii) injection rate—300 µl/min to 500 µl/min; and (iii) injection volume—50 µl.

Installing the control electronics for the ¹²⁹Xe MRI Set up the TTL output of the spectrometer with the photoMOS relay such that the latter controls the status of the shortcut valve (see **Fig. 5d** for the circuit). This is used to alternately force gas flow through the sample or to bypass the sample. From the Daq/board, use one analog output to control the set points for the flow and pressure controllers. Connect one analog input each to read the process values of the controllers. Display the corresponding parameters through appropriate elements in the

DASYLab/LabView software.

Preparation of the Xe gas-delivery manifold The timing of Xe bubbling is controlled by the pulse sequence and achieved through a proper gas-delivery system by connecting the PFA tubing and controllers according to **Figure 5e**. Before each experiment, insert the (five) capillaries into the capillary holder. The end tips should reach close to the bottom of the glass phantom. **▲ CRITICAL** This design of the flow and pressure controllers prevents Xe gas from passing through any needle valves or (electro)magnetic units during its transfer from the hyperpolarizer to the sample, thus minimizing the depolarization of Xe. **▲ CRITICAL** Gas bubbles can cause serious image artifacts if persistent during image acquisition. Therefore, the bubbling must be stopped and the sample must settle down before acquisition. Fresh hyperpolarized xenon must be re-delivered for each subsequent acquisition.

MRI pulse sequence setup The nonselective NMR sequence for acquiring direct Xe NMR spectra is adapted from an existing simple FID sequence by implementing the TTL trigger pulses and bubble/wait delays before the 90° block pulse (**Fig. 5e**). The FID sampling window is typically 1,000 ms and the spectral width to cover is ~350 p.p.m. For Xe MRI imaging sequence, single-shot pulse sequences should be used to make the most efficient use of the nonrenewable Xe polarization. A single-shot EPI pulse sequence can be used when long effective transverse relaxation time (T_2^*) values are expected. Alternatively, a single-shot RARE pulse sequence can be used and is limited by T_2 . To set up the imaging sequence, take a standard sequence with image encoding of your choice (EPI or RARE) and implement a magnetization transfer or saturation module (Bruker ParaVision terminology) after the TTL trigger pulse for xenon delivery and before the k-space encoding. The saturation offsets are defined through a saturation frequency list. This list can be accessed by providing in the ParaVision graphical user interface a menu to enter a fixed number of saturation pulse frequency offsets. Point these entries to the saturation frequency list. Finally, loop the entire xenon delivery, wait time, saturation pulse and imaging sequence multiple times to walk through the entire saturation frequency list. Some general parameters when implementing the sequence are as follows: a field of view: 20 × 20 mm²; matrix size: 32 × 32 (in-plane spatial resolution: 625 µm²); slice thickness: 20 mm; repetition time, including saturation time and xenon delivery: ~18 s.

Detailed parameters for RARE sequence: 90° Hermite excitation pulse (length = 3.375 ms, bandwidth = 1,600 Hz); 180° Mao refocusing pulse (length = 3.105 ms, bandwidth = 2,000 Hz); short echo time (~10 ms (this is also the effective echo time, as centric encoding is used here), echo readout time of 2.66 ms); Ensure that the RARE factor is set to give a single-shot acquisition, i.e., a RARE factor of 32 is used when the image matrix size is 32².

Detailed parameters for EPI sequence: 90° Gaussian-shaped excitation pulse: length = 1.0 ms, bandwidth = 2,740 Hz; encoding: partial Fourier-acceleration factor = 1.68 (i.e., 19 out of 32 k-space lines) and double sampling; short echo time of ~6 ms; echo train readout time of 6.01 ms for 19 k-space lines.

PROCEDURE

Production and purification of GV's

1| GV's can be produced natively in cyanobacterial (Ana GV's) or haloarchaeal (Halo GV's) hosts. In addition, they can be heterologously expressed in *E. coli* (Mega GV's). To produce and purify Ana, Halo or Mega GV's, follow the steps in option A, B or C, respectively.

(A) Production and purification of Ana GV's ● TIMING 4.5 weeks

- Creation of Ana starter cultures (Step 1A(i–iv): timing = 2 weeks).** Open the seal of the primary culture tube of Ana (as received from CCAP), and loosen the screw cap to allow aeration of the culture. Let the tube stand upright and undisturbed for 2 d in a cool (room temperature), dry place with access to ambient light. Viable Ana cells producing GV's will float to the top of the tube, forming a dark green layer.
- Take a presterilized Pyrex glass culture tube. Add 9 ml of G625 growth medium to the tube in the presence of a flame to ensure sterility. Add 0.2 ml of 50× BG-11 cyanobacteria freshwater solution. Swirl a few times to ensure complete mixing.
- Gently transfer 0.8 ml of the floating green cells from the primary culture to the G625 medium in the culture tube to make a total volume of 10 ml. Swirl gently to disperse the cells. The culture should have just a hint of green at this stage.

- (iv) Transfer the starter culture to the shaker incubator (shown in **Fig. 2b**) and allow it to grow at 25 °C, 100 r.p.m. and 1% CO₂ with cycles of 14 h of light illumination and 10 h in the dark until the cells form a dense, floating green layer that is clearly visible at the top of the liquid phase. This growth typically takes ~1–2 weeks. Note that the doubling time of Ana during exponential phase is ~56.5 h (ref. 25), so, depending on the number of viable cells in the starter culture, it will take 1–2 weeks to produce enough floating green cells that have the ability to produce GVs. Viable cells typically form the dense layer at the top and dead cells sink to the bottom of the tube.

? TROUBLESHOOTING

- (v) *Subculture of Ana for GV production (Step 1A(v and vi): timing = 2 weeks)*. Prepare aliquots of 200–250 ml of G625 medium in an autoclaved 1-liter Erlenmeyer flask (use a Bunsen burner flame or a laminar flow hood to provide a sterile environment to avoid contamination). Add a 1× final concentration of BG-11 freshwater solution (4–5 ml of the 50× stock) just before inoculation with Ana starter culture. Swirl to mix the components.
- (vi) Inoculate 2 ml of the green, floating layer of the starter culture per flask and grow cells in the incubator at 25 °C, 100 r.p.m., 1% CO₂, 14-h light and 10-h dark cycle until they become confluent (as shown in **Fig. 2b,c**). This should take ~2 weeks, with the exact time depending on the number of viable cells inoculated.

? TROUBLESHOOTING

- (vii) *Harvesting of Ana GVs (Step 1A(vii–xi): timing = 1 d)*. Pour the confluent culture from the flask into a separatory funnel (presterilized, with stopcock in place), allowing it to stand undisturbed on a metal ring stand for 12–16 h until the GV-producing Ana cells separate from the spent medium to form a compact green layer at the top (**Fig. 2e**).
- (viii) Carefully remove most of the spent medium by opening the stopcock at the bottom of the funnel and retaining only the green fraction at the top, along with ~30 ml of medium.
- (ix) Using a 25-ml serological pipette, gently resuspend the cells and transfer to a 50-ml tube.
- (x) Use 10 ml of fresh G625 growth medium to collect the cells sticking to the walls of the funnel. Each flask should be reconstituted to ~50 ml of concentrated cell suspension.

? TROUBLESHOOTING

- (xi) Split the suspension into two Falcon tubes and add an equal volume of 1 M sorbitol (final concentration is 500 mM) and 10% (vol/vol) Solulyse (i.e., 5.5 ml in 55 ml) to achieve hypertonic lysis. Place the tubes in a rotatory shaker at 4 °C for 6–8 h. Green lysates will have a milky tinge post lysis due to the presence of GVs.
- ▲ **CRITICAL STEP** We have observed that GVs become structurally compromised by certain detergents and surfactants used routinely for cell lysis such as BugBuster (EMD Millipore), Triton-X and SDS. It is therefore advisable to thoroughly characterize the effect of any new detergent on GV structure and integrity using techniques such as TEM and pressure absorbance spectroscopy before using them for ultrasound and MRI.

? TROUBLESHOOTING

- (xii) *Isolation and purification of Ana GVs from lysate (Step 1A(xii–xvii): timing = 2–3 d)*. Transfer 35 ml of cell lysate to a 50-ml tube and centrifuge at 350g at 4–8 °C for ~24 h in a table-top centrifuge, preferably equipped with a swinging-bucket rotor.
- #### ? TROUBLESHOOTING
- (xiii) Slowly remove as much as possible of the green subnatant liquid and cell pellet using a 25-ml serological pipette, without disturbing the white GV layer on top.
 - (xiv) For the second and third rounds of centrifugation (350g, 4–8 °C), resuspend the white supernatant layer containing GVs in 25 ml of sterile 1× PBS. During each resuspension step, gently wash the inner walls of the Falcon tube with fresh PBS to recover GVs sticking to the sides. The green subnatant should progressively become clearer with each spin. The second and third rounds of centrifugation are shorter than the first, with the endpoint being when all the GVs rise to the top of the tube.
 - (xv) For the final spin, resuspend the GVs in 10–12 ml of PBS (total volume can vary, depending on GV yield) and prepare aliquots in 2-ml tubes. Spin in a microcentrifuge at 350g for 4 h at 4–8 °C.
- ▲ **CRITICAL STEP** Do not fill the tube to the brim, as GVs will become stuck to the lid of the tube when they float to the top during centrifugation

? TROUBLESHOOTING

- (xvi) Using a 21.5-gauge flat needle attached to a 3-ml syringe, maintain the angle of the microcentrifuge tube and slowly insert the needle through the GV film to the bottom. Collect most of the subnatant and slowly take out the needle. Some GV loss will occur at this step. This is OK as long as most of the GVs are not sucked into the syringe.
- (xvii) Resuspend the purified GVs in ~6–8 ml of PBS (exact volume of added PBS can vary and is determined by how concentrated the GV solution must be for the end application) and prepare aliquots of the milky white GV solution (**Fig. 2j**) in screw-top vials or microcentrifuge tubes.

▲ **CRITICAL STEP** Avoid freezing or subjecting the tube to shocks, such as dropping it to the ground or snapping the cap, as these may collapse the GVs.

■ **PAUSE POINT** The aliquots of Ana GVs can be stored for up to a year at 4 °C.

(B) Production and purification of Halo GVs ● **TIMING 3.5 weeks**

- (i) *Growing of Halo cultures for GV production (Step 1B(i–iii): timing = 2 weeks).* Prepare 250-ml aliquots of Carolina growth medium into an autoclaved 1-liter Erlenmeyer flask under sterile conditions.
- (ii) Inoculate Halo cultures using one of the following methods. (i) Using a sterile pipette tip or toothpick, scrape a small amount of pink culture from the agar plate to add to the flask as inoculum. (ii) Use one to two brine crystals containing Halo for inoculation. (iii) Inoculate 2.5 ml from a healthy pink liquid starter culture into 250 ml of fresh growth medium (1:100).
- (iii) Grow the culture in an incubator at 42 °C with shaking at 100 r.p.m. Depending on the health of the parent culture, it may take ~2 weeks for the inoculated culture to become confluent. Confluency is determined by the color and turbidity of cultures, as shown in **Figure 2d**.

? TROUBLESHOOTING

- (iv) *Harvesting of Halo GVs (Steps 1B(iv–vi): timing = 1 week).* Gently pour the culture from the flask into a separatory funnel (presterilized, with stopcock in place). Allow the culture to remain undisturbed until a visible ring is formed at the top (**Fig. 2f**). This typically takes 4–6 d.

? TROUBLESHOOTING

- (v) Remove as much of the spent medium as possible by opening the stopcock, retaining only the buoyant layer of milky pink cells for lysis. The retained volume is ~10–20 ml, and many cells stick to the sides of the funnel.
- (vi) Using an equal volume (10–20 ml) of TMC lysis buffer (pH 7.5), gently wash the cells stuck to the sides of the funnel and retrieve as many cells as possible. The volume of TMC buffer used may be varied according to the cell density, in order to achieve efficient hypo-osmotic lysis. However, note that if too much buffer is used in this step, the number of 2-ml aliquots will proportionately increase for the next step, thus increasing sample processing time.

? TROUBLESHOOTING

- (vii) *Isolation and purification of Halo GVs from lysate (Step 1B(vii–xi): timing = 2 d).* Prepare ~1.6-ml aliquots of cells in 2-ml tubes and spin in a microcentrifuge at 300g for 4 h at 4–8 °C.

▲ **CRITICAL STEP** Close the tubes gently; the pressure wave caused by snapping the lid will collapse a large number of the Halo GVs.

? TROUBLESHOOTING

- (viii) At the top of the tube, a mixed layer of Halo GVs (white) and unlysed Halo cells (milky pink) will be visible. Using a blunt-end 18.5- or 21.5-gauge needle, aspirate the pellet at the bottom of the tubes, as well as the pink cell lysate.

▲ **CRITICAL STEP** Take care to limit the amount of floating Halo cells and Halo GVs (white) that are aspirated into the syringe.
- (ix) Transfer the GVs and unlysed Halo cells to fresh tubes and bring the volume to 1.6 ml with 1× PBS. Centrifuge tubes at 300g for 4 h at 4–8 °C.
- (x) Repeat steps viii and ix. After each step, the amount of milky pink buoyant cells will be reduced and the amount of white Halo GVs will increase. Continue with centrifugally assisted floatation until all the cells have lysed and there is no evidence of pink cell lysate in the supernatant.
- (xi) Resuspend the purified GVs in PBS and prepare aliquots of the milky white GV solution (**Fig. 2j**) in screw-top vials or microcentrifuge tubes.

▲ **CRITICAL STEP** Avoid freezing and subjecting the tube to mechanical shocks, such as dropping it to the ground or snapping the cap, as these may collapse the GVs.

■ **PAUSE POINT** The aliquots of Halo GVs can be stored for up to one year at 4 °C.

(C) Production and purification of Mega GVs ● **TIMING 4 d**

- (i) *Heterologous expression of Mega GVs in E. coli (Step 1C(i–iv): timing = 2 d).* Transform 50 µl of chemically competent Rosetta 2(DE3) pLysS cells using >1 ng of pST39 plasmid containing the pNL29 Mega GV gene cluster¹⁵ by mixing the two components in a 1.5-ml tube and incubating on ice for 30 min. Heat-shock the tube in a 42 °C water bath for 45 s, and put the tube back on ice for 1 min. Add 500 µl of SOC outgrowth medium and incubate in a shaker at 37 °C and 250 r.p.m. for 1 h.
- (ii) Prepare 3 ml of LB medium containing 1× ampicillin (100 µg/ml), 1× chloramphenicol (25 µg/ml) and 1% (wt/vol) glucose in a glass culture tube. Resuspend 300 µl of the transformed *E. coli* in the broth. Grow the culture in a shaker incubator at 37 °C and 250 r.p.m. until the OD₆₀₀ value reaches 0.4–0.6. Make 100-µl aliquots of the culture in sterile tubes, and mix with 100 µl of 50% sterile glycerol. Freeze the tubes at –80 °C as *E. coli* glycerol stocks.

■ **PAUSE POINT** The glycerol stocks can be stored at –80 °C and used for up to 3 months. Note that although using glycerol stocks is convenient, the GV yield is reduced when using frozen stocks, so we recommend fresh overnight transformations for best results.
- (iii) Resuspend an aliquot of glycerol stock in 3 ml of LB medium containing 1× ampicillin, 1× chloramphenicol and 1% (wt/vol) glucose. Grow the *E. coli* culture to saturation (OD₆₀₀ > 4). For fresh transformations, add 500 µl of

the transformed *E. coli* from Step 1C(i) to 5 ml of LB medium containing 1× ampicillin, 1× chloramphenicol and 1% (wt/vol) glucose. Allow the culture to grow overnight until it reaches saturation (~16 h).

- (iv) Prepare 100 ml of LB medium containing 1× ampicillin, 1× chloramphenicol and 0.2% (wt/vol) glucose, and inoculate 1 ml of the saturated *E. coli* culture into the broth. Grow at 37 °C for ~2 h until the OD₆₀₀ value reaches 0.4 to 0.6. Induce the culture by adding 20 μM IPTG (final concentration), and grow it at 30 °C for an additional 16–24 h.
- (v) *Harvesting and purification of Mega GVs from E. coli cultures (Step 1C(v–x): timing 2 d)*. Split the culture equally into three 50-ml Falcon tubes and spin for 1 h at 500g and 25 °C.

▲ **CRITICAL STEP** Avoid higher speeds because they may cause collapse of GVs.

? TROUBLESHOOTING

- (vi) Insert a 10-ml syringe with a needle to >1 cm below the surface of the solution and withdraw the clear liquid component of the solution. Withdraw the liquid slowly to preserve the thin layer of cells floating at the top of the solution, as well as the pellet at the bottom, both of which contain Mega GVs. Resuspend cells in the remaining media and use 1–5 mL of 1× PBS to retrieve cells sticking to the sides of the tube. Pool cells from all the tubes in a fresh Falcon tube.
- (vii) To lyse the cells, add 5 ml of SoluLyse–Tris reagent per 50 ml of pooled *E. coli* culture, 250 μg/ml lysozyme and 10 μg/ml DNaseI. Rotate the tubes for 10 min at room temperature and then prepare 1.5-ml aliquots of the solution in 2-ml tubes. Spin samples for 4 h at 800g and 4–8 °C. Mix the floating GV layer gently with the supernatant and transfer to a clean tube.
- (viii) Spin the samples for 4 h at 800g and 4–8 °C. Use a 3-ml syringe to remove the bottom fraction, which sometimes includes a small pellet. Gently resuspend GVs in 1 ml of PBS. Repeat the spin and wash steps 3 times.

▲ **CRITICAL STEP** Be aware that GVs are susceptible to desiccation, and resuspend GVs immediately after withdrawing the liquid.

- (ix) Mega GVs are natively clustered. To uncluster them, GV-containing solution is mixed with 10 M urea in a 2:3 ratio to achieve a 6 M final urea concentration, and the resulting solution is gently rotated for 30 min.
- (x) (Optional) Dialyze GVs overnight in 6–8 kDa MWCO tubing against 4 liter of PBS. This step can be omitted for experiments with no stringent requirements for buffer conditions. The white layer of unclustered GVs at the top of the liquid phase after buoyancy purification, as well as the resuspended milky white solution of Mega GVs in PBS, is shown in **Figure 2i,j**.

▲ **CRITICAL STEP** Avoid freezing and subjecting the tube to mechanical shocks, such as dropping it to the ground or snapping the cap, as these may collapse the GVs.

■ **PAUSE POINT** Mega GVs can be stored for up to one year at 4 °C.

Quantification of GVs by measurement of pressure-sensitive OD ● **TIMING 15 min**

2| Determine the concentration of a solution of GVs by measuring its pressure-sensitive OD at 500 nm (OD_{500,ps}) using a NanoDrop 2000 spectrophotometer. Load 2 μl of sample onto the pedestal for each measurement. Collapsed GVs in the same buffer are used as a blank for measurements. Prepare collapsed GVs by sonication in a water bath until the solution turns completely clear or by manual collapse in a capped syringe. For manual collapse, remove the plunger from a 12-ml Luer-Lock syringe closed with a tip cap and place 5–10 μl of GV solution at the bottom of the syringe. Making sure that the tip cap is screwed on tight, replace the plunger and push down until there is substantial resistance. The increase in pressure will collapse the gas vesicles, turning the milky white solution clear. A shortcut for quick measurements of GV concentration is to use the GV resuspension buffer as a blank. For most samples, this will yield an OD reading that is very close to that measured when using collapsed GVs as a blank. However, for some samples containing GVs that are fluorescent, it is necessary to use the collapsed GVs as the blank.

! **CAUTION** Before performing manual collapse, ensure that the tip cap on the syringe is facing away from you and is securely fastened, as pressurization of the syringe may cause ejection of the cap, resulting in potential injury to the user. Wear protective eyewear and clothing.

▲ **CRITICAL STEP** It is important to ensure that the GVs are homogeneously resuspended in solution just before measurements. For each sample, take the average OD_{500,ps} value after multiple measurements ($n > 3$) to ensure precision and accuracy.

Quantification of GV protein concentration ● **TIMING 15 min**

3| Measure the protein concentrations using the Pierce 660-nm protein assay to obtain relationships between OD and protein content for the GV solutions. We have established the protein concentration to OD relationships for our three types of GVs, and the results are as shown in **Table 3** below ($N = 4, 5$ and 3 for Mega, Ana and Halo GVs, respectively, and the errors are SEMs). The molecular weight is derived from the TEM data summarized in **Table 4**, assuming a spindle shape for Halo GVs, a cylindrical shape for Mega and Ana GVs, a wall thickness of 18 Å and a protein density of 1.4 g/ml.

TABLE 3 | Quantification and calculation of GV molecular weight and molar concentration.

	Ana	Halo	Mega
Protein concentration to OD ₅₀₀ ratio ((μg/ml)/OD)	36.6 ± 2.6	13.4 ± 2.2	145.5 ± 6.4
Estimated molecular weight (MDa)	320	282	71.7
Estimated molar protein concentration to OD ₅₀₀ ratio (pM/OD)	114	47.3	2,030
Estimated gas fraction to OD ₅₀₀ ratio (vol/vol/OD)	0.000417	0.000178	0.000794

Chemical functionalization of GVs ● TIMING 2 d

▲ **CRITICAL** Purified Ana, Halo and Mega GVs contain lysine residues on the surface that can be used to chemically conjugate a variety of moieties, such as PEG, fluorophores and biotin, using an amine-reactive coupling group such as *N*-hydroxy-succinimide ester.

4| Measure the concentration of purified GVs using the OD relationships in **Table 3**.

5| Prepare aliquots of the NHS moiety in anhydrous DMSO at 100× the required molar concentration for the amine-NHS reaction. For Alexa-488-NHS conjugation to Ana GVs, add 5 μl of the 10 mM stock solution of the dye predissolved in anhydrous DMSO.

▲ **CRITICAL STEP** Ensure that the NHS moiety solution does not contain detergents or surfactants that might affect the integrity and properties of the GVs.

6| Adjust the concentration and volume of GVs to the desired amount and ensure that the buffer is free of amines (avoid Tris buffer). For Alexa-488-NHS conjugation to Ana GVs, bring Ana GVs to an OD value of 1 in 1 ml of PBS at pH 7.4.

▲ **CRITICAL STEP** If GVs were previously in a buffer containing free amines or PBS with a pH <7, ensure complete buffer exchange with PBS at a pH of 7–9 before proceeding with the amine-NHS reaction.

7| Add a 10⁵ molar excess of the NHS moiety to the GVs, keeping the DMSO concentration 0.5% of the total reaction volume. For Alexa-488-NHS conjugation to Ana GVs, add 5 μl of the 10 mM Alexa-488-NHS in DMSO to 1 ml of Ana GV solution. On the basis of the average number of GvpA and GvpC protein monomers that make up Ana GVs, ~50,000 lysine residues are present for each Ana GV. One can tune the molar ratio of the two reactants (NHS moiety/GV) to achieve the desired reaction efficiency.

8| Allow the reaction to proceed for 4 h at room temperature under gentle rotation.

? TROUBLESHOOTING

9| Quench the unreacted NHS moieties using Tris-HCl buffer at pH 8 to a final concentration of 10 mM for 20 min at room temperature under gentle rotation

10| Add the whole reaction mixture to dialysis tubes (6–8 kDa cutoff) and dialyze against a 4,000× volume excess of PBS at 4 °C for 8 h. Replace the buffer and allow dialysis to continue for an additional 8 h.

▲ **CRITICAL STEP** If the NHS moiety is not amenable to dialysis, the use of repeated rounds of centrifugally assisted purification is an alternative method to remove excess reactants and/or for buffer exchange.

? TROUBLESHOOTING

■ **PAUSE POINT** Chemically functionalized GVs can be stored in PBS buffer for up to 1 year at 4 °C.

(Optional) Genetic modification and functionalization of Ana GVs ● TIMING 3 d

▲ **CRITICAL** The outer scaffold protein of Ana GVs, GvpC, can be removed and replaced with genetically modified recombinant versions as a molecular handle for GV mechanical modification and functionalization. This procedure comprises stripping native GvpC from Ana GVs (Steps 11–19), preparation of recombinant GvpC (Steps 20–38) and re-adding this GvpC onto ΔGvpC Ana GVs via dialysis (Steps 39–41). In addition, GVs functionalized with GvpC fused to the SpyTag peptide enable convenient downstream covalent functionalization with proteins fused to the SpyCatcher moiety (Steps 42–45).

11| Preparation of ΔGvpC GVs (Steps 11–19: timing = 12 h). Dilute purified Ana GVs in PBS such that the OD_{500,ps} value <10.

12| Prepare a 3:2 (vol/vol) mix of GV stripping buffer (Round 1; i.e., 10 M urea in 100 mM Tris buffer) and GV solution in PBS. Pipette 1.7 ml into 2-ml microcentrifuge tubes.

13 | Centrifuge at 300g for 4 h at 4–8 °C or until the supernatant is completely clear. Remove the clear supernatant with a syringe using a 21-gauge flat needle. Retain the milky white supernatant in the tube. Resuspend the GV-containing supernatant in GV stripping buffer (Round 2; i.e., 6 M urea, 60 mM Tris-HCl). Repeat this step once.

14 | Confirm GvpC removal by performing SDS–PAGE. Incubate a 1:1 (vol/vol) mix of GVs in 2× Laemmli buffer (containing 5% (vol/vol) 2-mercaptoethanol) at 95 °C for 5 min. Centrifuge briefly to collect condensate.

15 | Remove the comb and bottom tape from the precast polyacrylamide gel. Assemble the electrophoresis cell according to the manufacturer's instructions. Fill the inner chamber completely with 200 ml of 1× TGS buffer. Ensure that the inner chamber is not leaking fluid. Fill the outer chamber up to the mark indicated on the buffer tank with 600 ml of 1× TGS buffer.

16 | Load the protein ladder and samples into the gel using gel-loading tips. GVs should be at OD₅₀₀ > 3 before the 1:1 dilution. If purified proteins are being run on the same gel for comparison, load >100 ng.

▲ **CRITICAL STEP** To prevent contamination between wells, do not exceed the maximum recommended volume per well.

17 | Connect the electrophoresis cell to the power supply and run the gel for 55 min at 120 V.

18 | Recover the gel by disassembling the electrophoresis cell and the gel cassette. Incubate the gel in a holder with ddH₂O for 10 min, and then stain for 1 h with 10 ml of SimplyBlue SafeStain. Destain the gel for at least 1 h with 10 ml of ddH₂O.

19 | Image the gel using a Coomassie imaging protocol, using the gel imaging system to visualize protein bands. The GvpC band at ~25 kDa should be missing.

? TROUBLESHOOTING

■ **PAUSE POINT** Store the ΔGvpC GVs in urea buffer at 4 °C for no more than 1 week. When preparing ΔGvpC GVs for long-term storage without any further genetic functionalization or recombinant GvpC addition (Steps 39–45), we recommend dialyzing the GV solution against PBS in order to completely remove the urea.

20 | *Preparation of recombinant GvpC (Steps 20–38: timing = 1 d).* Transform >1 ng pure plasmid encoding recombinant GvpC with a C-terminal hexahistidine tag into BL21 (DE3) competent cells and grow the culture in terrific broth with 50 μg/ml kanamycin overnight.

21 | Dilute 500 μl of the starter culture 1:1 with 50% glycerol in water and store it at –80 °C for up to one year for the best results. Future starter cultures can be grown from aliquots of this glycerol stock instead of fresh transformations.

22 | Dilute the starter culture 1:250 in terrific broth with 50 μg/ml kanamycin and grow it with shaking (250 r.p.m.) at 37 °C until the OD₆₀₀ value is ~0.4–0.7. Induce at a final concentration of 1 mM IPTG. Grow the culture for 6–12 h at 30 °C with shaking at 250 r.p.m.

▲ **CRITICAL STEP** Frozen IPTG stocks should be thawed fully and vortexed to mix contents before use.

23 | Pellet the cells in ultracentrifuge tubes at 5,500g for 15 min at 4 °C and discard the supernatant.

■ **PAUSE POINT** Cell pellets can be stored at –20 °C for up to one year. Protein extraction is typically more effective with frozen cells.

24 | Resuspend the pellets (typically from 150 ml of liquid culture) in 10 ml of Solulyse with 10 μg/ml DNase. Rotate at room temperature for 10 min.

25 | Centrifuge at 20,000g for 15 min at 4 °C to clear the lysate and discard the supernatant.

26 | Resuspend the pellet in 10 ml of Solulyse and lysozyme (0.25 mg/ml). Rotate at room temperature for 10 min.

27 | Add 5 ml of Solulyse and vortex. Centrifuge at 20,000g for 20 min at 4 °C and discard the supernatant.

TABLE 4 | GV dimensions.

Spatial dimension	Ana GV	Halo GV	Mega GV
Length (nm)	519 ± 160	400 ± 113	249 ± 99
Width (nm)	136.3 ± 21.0	250.8 ± 51.4	72.5 ± 13.6

N = 107, 125 and 61 for Ana, Halo, and Mega respectively; errors indicate standard deviation.

PROTOCOL

28| Thoroughly resuspend the pellet in 10 ml of inclusion body solubilization buffer. The exact volume of solubilization buffer can be adjusted according to the amount of expression culture and the level of protein expression (i.e., based on the size of the white inclusion body pellet) in order to achieve effective resuspension. Typically, 10 ml of buffer is used for a pellet from 150 ml of expression culture.

29| Centrifuge at 30,000g for 20 min at 4 °C.

30| Add 1.5 ml of Ni-NTA slurry to the supernatant, and incubate at 4 °C with shaking (60 r.p.m.) for 2 h or more.

31| Pour into a Poly-Prep column and collect all the flow-through, wash and elutions in the next steps. Collecting all fractions is good practice for troubleshooting and analyzing purification steps using SDS-PAGE.

32| Wash with 10 column volumes of inclusion body wash buffer.

33| Elute with 2 column volumes of inclusion body elution buffer to collect Elution 1. Repeat once to collect Elution 2.

34| To quantify the eluted protein using the Bradford assay, prepare a standard curve of BSA at final concentrations of 100, 250, 500, 750, 1,000 and 1,500 µg/ml in 60 µl of 3× diluted inclusion body elution buffer in PBS. Prepare dilutions of eluted protein 1:2 in PBS with a final volume of 60 µl. Prepare a negative control of 3× diluted inclusion body elution buffer in PBS.

35| To 25 µl of the sample and BSA standards, add 1 ml of Bradford reagent, vortex and incubate at room temperature for 5–10 min. Prepare all samples in duplicate.

36| Blank the spectrophotometer with a negative control sample and measure the OD₅₉₅ value.

37| Measure the OD₅₉₅ value of the standard curve samples. Plot the OD₅₉₅ value versus the concentration, and compute linear regression fit.

38| Measure the OD₅₉₅ value of the eluted protein samples. Use the linear fit from Step 37 to compute the unknown concentrations, and multiply by 3 (dilution factor) to obtain the concentration of the stock elution solution.

? TROUBLESHOOTING

■ **PAUSE POINT** Store Elutions 1 and 2 separately at 4 °C for up to 3 months. Elution 1 has > 80% of collected pure protein and is used for the subsequent experiments. Elution 2 is more dilute and is typically stored as backup or for running protein controls for SDS-PAGE.

39| *Preparation of GVs with recombinant GvpC (Steps 39–41: timing = 1 d).* Add recombinant GvpC to ΔGvpC GVs according to the formulation: $2 \times \text{OD} \times 198 \text{ nM} \times \text{volume (in liters) of GVs} = \text{nmol of recombinant GvpC}$. This provides a twofold stoichiometric excess of GvpC relative to binding sites on an average Ana GV, assuming a 1:25 molar ratio of GvpC/GvpA binding based on previous work²⁶. The exact volume of recombinant GvpC to be added is calculated based on the molar mass of the particular variant and the concentration of eluted GvpC (measured by Bradford assay according to Steps 34–38). For truncated GvpC variants with a lower GV binding affinity, a higher stoichiometric excess may be added to promote attachment of GvpC to the GV surface. However, note that adding too much excess of GvpC might lead to protein aggregation during dialysis.

40| Soak the dialysis tubing in PBS for 5 min. Add samples (GVs + recombinant GvpC) into dialysis tubing and clip both sides. Dialyze in 4 liters of PBS with stirring on low speed for at least 12 h.

▲ **CRITICAL STEP** The length of dialysis tubing used for each sample depends on the total volume of the dialysate, which is determined by the amount of engineered GVs required for the end application. The type of dialysis tubing used (molecular weight cutoff) can change depending on the GvpC variant, as truncated variants may have a much lower molecular weight.

41| Transfer the dialysate into 2-ml centrifuge tubes and spin at 4–8 °C at 300g for 3 h, or until the supernatant is clear. Remove the supernatant with a syringe with a 21.5-gauge flat needle. Retain the milky white supernatant in the tube. Resuspend GVs in PBS. Repeat this centrifugation step once.

? TROUBLESHOOTING

■ **PAUSE POINT** Store at 4 °C for up to 3 months. Note that as different GvpC variants have different binding affinities to the GvpA shell, it is advisable to check the hydrostatic collapse profile of the stored GV variant before use in imaging experiments, to ensure that there is no dissociation of recombinant GvpC from the GV surface during storage.

42| (Optional) Preparation of SpyCatcher-functionalized GVs (Steps 42–45: timing = 5 h). Prepare Ana GVs with GvpC fused to the SpyTag peptide using Steps 39–41.

43| Mix SpyTag-functionalized GVs with SpyCatcher-fused proteins according to the following formulation: $2 \times \text{OD} \times 396 \text{ nM} \times \text{volume (in liters) of SpyTag GVs} = \text{nmol SpyCatcher-fused protein}$. This results in a twofold excess of SpyCatcher to SpyTag in the reaction, based on the stoichiometry described in Step 39. Note that the SpyCatcher-mNeonGreen (SC-mNG) fusion protein used in our published work¹⁴ is expressed separately in *E. coli* following procedures in Steps 20–23 and using the plasmid containing SC-mNG (details in the Reagents section). SC-mNG is expressed as a soluble and hexahistidine-tagged protein, enabling purification using the same Ni-NTA slurry used for recombinant GvpC purification. Unlike GvpC inclusion bodies, soluble proteins are in the supernatant after cell lysis with Solulyse/DNase (Step 24), allowing direct incubation of the supernatant with the slurry (Step 30). Wash and elution (Steps 31–33) are performed with soluble protein wash buffer and soluble protein elution buffer, respectively, and the protein is desalted into PBS using PD10 desalting columns. Protein quantification is done using the Pierce or Bradford assay before use.

44| Incubate for 1 h or more at room temperature.

45| Centrifuge at 300g for 3 h at 4–8 °C or until the supernatant is clear. Remove the clear supernatant with a syringe with a 21.5-gauge needle. Retain the milky white supernatant in the tube. Resuspend the supernatant in PBS. Repeat this centrifugation step one time.

? TROUBLESHOOTING

■ **PAUSE POINT** Store the GV suspension at 4 °C for up to 3 months.

GV characterization

46| To characterize the purified GVs, follow the procedure in option A to determine the critical collapse pressure of GVs using pressurized absorbance spectroscopy. Follow option B for DLS measurements and option C for preparing the GV specimens for TEM.

(A) Determination of critical collapse of GVs with pressurized absorbance spectroscopy ● TIMING 40 min

- (i) Before acquiring measurements, connect the spectrophotometer to a power supply and turn it on 30 min before taking measurements to allow the light source to warm up and stabilize.
- (ii) Run the Alicit_startup script (**Supplementary Data**) to initialize the pressure controller.
- (iii) Blank the spectrophotometer with a cuvette filled with PBS or GV resuspension buffer and run OceanOptics_startup_FullTrans (**Supplementary Data**) to save the data.
- (iv) Establish a zero-transmission baseline with the opaque side of the cuvette blocking the light path using the OceanOptics_startup_NoTrans script (**Supplementary Data**).
- (v) Fill the cuvette with intact GV sample ($\text{OD}_{500\text{nm}} = 0.2$ in PBS) and fasten the cannulae securely. To assist loading, use elongated gel-loading micropipette tips.
- (vi) Open the N_2 tank valve, pressure regulator and pressure controller valve.
- (vii) Run the Collapse_script (**Supplementary Data**) to measure the $\text{OD}_{500,\text{ps}}$ value under increasing hydrostatic pressure (0 kPa—1.4 MPa in 20-kPa increments).
- (viii) Between these measurements, rinse the cuvette with ddH₂O, 70% ethanol and acetone to ensure that the cuvette is completely clean and dry before adding the next sample.
- (ix) After measurements, close the gas valves and turn off the spectrophotometer.

? TROUBLESHOOTING

(B) Preparation of GV specimens for DLS ● TIMING 5 min

- (i) Dilute the GV samples to $\text{OD}_{500,\text{ps}} = 0.2$ in PBS.
- (ii) Measure the particle size on the ZetaPALS instrument with an angle of 90 ° and refractive index of 1.33.

(C) Preparation of GV samples for TEM ● TIMING 30 min

- (i) Buffer-exchange the purified GVs in 10 mM HEPES buffer with 150 mM NaCl or an alternative non-phosphate-containing buffer via centrifugally assisted floatation (same procedure as used for GV isolation) at 4–8 °C and 300g. Replace the supernatant with an equal volume of 10 mM HEPES buffer with 150 mM NaCl. Repeat this step three times. The aim of this step is to prevent phosphate in the PBS from causing unwanted precipitation of the uranyl acetate stain used downstream in Step 46C(vi). Therefore, if the GV solution is very concentrated, direct dilution of the GV sample into the above-mentioned HEPES buffer to a final OD of 0.2 would be a quicker alternative.
- (ii) Dilute the GVs to a final OD value of 0.2.
- (iii) Spin 2% uranyl acetate solution in a benchtop centrifuge at 14,000g for 5 min at room temperature to pellet any precipitate.
- (iv) Using the glow discharge system, charge Formvar TEM grids with 15 mA current for 1 min.
▲ **CRITICAL STEP** Gently handle the TEM grids and avoid contact of the tweezer tip with the center of the grid to prevent bending and damage to the grids.

PROTOCOL

- (v) Place 2 μl of well-mixed GV solution on the charged Formvar TEM grids for 3 min. The sample should be placed on the carbon side of the grid; avoid placing sample on the copper side. For convenience, we use Pelco reverse, anti-capillary tweezers to hold the TEM grid while adding sample and negative stain.
- (vi) Add 5 μl of 2% uranyl acetate to the GV solution on the TEM grid for 30 s.
- (vii) Using a Whatman filter paper, wick the solution by gently touching the edge of the grid. For consistent results, leave a thin film of sample on the grids and allow it to air-dry.
- (viii) Image the grids using TEM.

? TROUBLESHOOTING

Ultrasound imaging of GVs *in vitro* in agarose phantoms

▲ **CRITICAL** Steps 47–63 describe the application of GVs as contrast agents for ultrasound. For applications in MRI, proceed to Step 64.

47| *Agarose phantom preparation and sample loading (Steps 47–55: timing = 3 h).* Make a 250-ml solution of 1% (wt/vol) agarose in PBS and microwave for 3 min to ensure dissolution. Ensure that the lid is loosely fastened and swirl the bottle at 1-min intervals to prevent the agarose from settling.

48| Fasten the lid and put the bottle in a water bath at 50–55 °C for at least 30–45 min to allow the solution to equilibrate to the bath temperature

49| Fill the Nalgene plastic holder (12.5-cm length $\times$ 8-cm width, 300 ml) with 150 ml of molten 1% (wt/vol) agarose in PBS. Keep the remaining agarose solution in the water bath for sample loading.

50| Immerse the multiwell phantom mold on top of the agarose solution.

▲ **CRITICAL STEP** Remove any small air bubbles with a syringe needle and lower the mold gently onto the surface of the agarose, maintaining a small angle to allow any air trapped on the interface between the agarose and mold to escape.

51| Let the agarose phantom solidify at room temperature for 1 h.

52| Carefully remove the multiwell phantom mold without disturbing the agarose beneath.

53| Prepare the GV samples and controls at 2 $\times$ the final desired concentration.

54| Prepare a 1:1 mixture of the 1% (wt/vol) agarose in PBS with the GVs or polystyrene beads.

▲ **CRITICAL STEP** Make sure that the 1% (wt/vol) agarose in PBS is equilibrated to a temperature of 50–55 °C before and during the mixing with GVs or polystyrene beads. Just before mixing with the agarose, place the GV or polystyrene sample for 10–15 s in a 50 °C water bath to warm the sample. This allows proper mixing of agarose with the sample and provides enough time to load the sample into the well before the agarose solidifies.

▲ **CRITICAL STEP** Do not heat the GV samples to temperatures >50 °C. GV collapse has been observed even for a 10- to 20-s incubation at 65 °C.

55| Pipette the mixture into the phantom well. The volume of mixture pipetted depends on the dimensions of the well, typically 50–100 μl .

56| *Phantom imaging (Steps 56–62: timing = 1 h).* Place the ultrasound acoustic absorber at the bottom of the phantom holder to minimize multiple ultrasound wave reflections during the measurement.

57| Immerse the phantom in water or 1 $\times$ PBS.

58| Perform ultrasound imaging on the Verasonics L11-4v ray-line example script (provided by the manufacturer) using the following parameters: set the transmit frequency to 4.5 MHz, set the number of cycles of the transmitted ultrasound pulse to 3, the F number to 3, the voltage to 2.5 and the persistence to 90. These parameters can be adjusted by the user in the Verasonics MATLAB script defining the ultrasound pulse sequence, and operated via the VSX GUI. Perform the collapse using a high-power pulse at 25 V.

59| Perform ultrasound imaging at the fundamental frequency with the L22-14v: set the transmit frequency to 11.5 MHz, the number of cycles of the transmitted ultrasound pulse to 6, the F number to 2 and the voltage to 3 V. Those parameters are adjusted in the script and operated via the VSX GUI.

60 Perform ultrasound imaging at the second harmonic frequency with the L11-4v transducer: echoes at the second harmonic frequency are displayed using a 2-MHz-wide band-pass filter centered ~9 MHz. Other parameters are identical to those of the fundamental imaging mode and adjusted in the script. Second harmonic imaging is operated as an option via the VSX GUI. Note that we have recently developed an amplitude modulation strategy that can also be used for highly selective nonlinear imaging of native and engineered GVs⁹.

61 Obtain acoustic multiplexing images by using GVs at OD = 1, a transmit frequency of 6.25 MHz and a voltage of 1.6 V on the L11-4v, with a four-cycle pulse, transmit focus of 20 mm, F number of 2 and persistence of 90. To collapse the GVs, increase the acoustic pressure by lowering the F number to 0.1 and gradually ramping up the voltage. At each collapse step, the transducer is translated in the y and z planes using the Velmex motors to ensure homogeneous collapse over the entire well and field of view.

62 Save the ultrasound images for subsequent processing as .fig images (MATLAB format) directly from the MATLAB figure panel.

7 TROUBLESHOOTING

Ultrasound imaging of GVs *in vivo*

63 GVs can be used as contrast agents for *in vivo* ultrasound imaging. Option A describes the steps for imaging GVs in mice using the Verasonics system, whereas option B describes the procedure for imaging Halo GVs using the Vevo ultrasound system.

! CAUTION Any experiments involving live mice must conform to relevant institutional and national regulations. All animal experiments reported in this article received approval from the institutional animal care and use committees of the California Institute of Technology or the Sunnybrook Research Institute.

(A) GV imaging *in vivo* using the Verasonics ultrasound system ● TIMING 45 min

- (i) Place the mouse in the induction chamber with 3–4% isoflurane, room air at 2 liters/min with the vacuum on for isoflurane removal.
- (ii) When the animal is unresponsive to toe pinch, move the animal to the heat pad and nose cone for continued anesthesia and place it in supine position. Reduce the isoflurane to 2% and apply eye lubricant to both eyes to avoid drying. Use Nair and cotton swabs to depilate the abdominal area and wash it thoroughly with saline to avoid any burns on the skin.
- (iii) Prepare the catheter using 10 cm of PE10 plastic tubing and a 30-gauge needle. With a razor blade, cut the plastic at the bottom of the needle to allow the blunt end to be free. Push this end into the PE10 plastic tubing.
! CAUTION Follow safe practices and wear proper personal protective equipment while handling sharp needles.
- (iv) Fill the catheter with saline using a 30-gauge syringe. Confirm that there is no leaking from the catheter. Remove the saline-filled syringe before the next step.
! CAUTION Failure to fill the tubing with liquid to remove the air from the catheter can lead to a gas embolism and death of the animal.
- (v) To insert the catheter into the tail vein, place the mouse on its side for a better view of the lateral tail vein and clean the tail with an alcohol swab. Use the dominant hand to hold the catheter; use the other hand to hold the tail between your thumb and index finger. Insert the bevel needle through the skin and into the vein at a flat angle and advance it for several millimeters. If the needle is inserted correctly, blood should backflow through the catheter.
- (vi) Secure the position of the catheter in the tail vein with tissue adhesive glue and use the 30-gauge syringe filled with saline to flush the blood from the catheter.
▲ CRITICAL STEP The flow of saline should be smooth, confirming proper positioning of the needle tip in the tail vein. Excessive pressure to achieve flow may indicate improper positioning of the catheter or clotting of the blood.
- (vii) To perform ultrasound imaging, slowly adjust the mouse to a supine position and apply ultrasound gel on the depilated area. Place the transducer on the gel.
▲ CRITICAL STEP Make sure that there are no air gaps and that the gel is completely coupled to the transducer.
- (viii) Open the Verasonics L11-4v ray-line example script (provided by the manufacturer) in MATLAB. The transmit frequency is set to 4.5 MHz with the L11-4v transducer. Position the transducer to see the inferior vena cava (IVC) for vascular imaging. Keep the settings at 2.5 V, 3-cycle pulse, F number 3 and persistence 20.
▲ CRITICAL STEP Ensure that the region of interest in the mouse aligns with the focus of the transducer to acquire optimal images.
- (ix) Remix the gas vesicles thoroughly by gently pipetting up and down, and then add 100 µl of the GV solution at OD_{500,ps} ~25 to a 1.5-ml microcentrifuge tube.
▲ CRITICAL STEP Gas vesicles must be mixed thoroughly each time before an injection; they are buoyant and will float up to the top of the tube. Make sure to inject immediately after remixing.

- (x) Carefully load a 30-gauge tuberculin syringe with the GV solution, making sure that there are no air bubbles. Do not tap, as this may collapse the GVs. Replace the saline-filled syringe at the end of the catheter and secure it to the syringe pump.
- (xi) Fill the catheter line with GVs at a rate of 5 $\mu\text{L/s}$.
- (xii) Inject 50 μL of GVs into the tail vein at a rate of 5 $\mu\text{L/s}$, and record the ultrasound signal appearing in the IVC and liver.
▲ CRITICAL STEP The total amount of injected saline and sample should be monitored and within the approved safety guidelines to prevent adverse effects on the animal caused by blood dilution.
- (xiii) Perform imaging using B-mode or harmonic filtering as described in Steps 58–60 for *in vitro* specimens.
- (xiv) At the end of the imaging session, turn off the isoflurane and remove the mouse from the nose cone. Wait for the animal to completely recover from the anesthesia before returning it back to the cage.

? TROUBLESHOOTING

(B) Halo GV imaging *in vivo* using the Vevo ultrasound system ● TIMING 45 min

- (i) Set up the ultrasound mouse imaging station (see ‘Setup of the Vevo Imaging Station’ in Equipment Setup) and attach the 21-MHz, MS250 MicroScan Transducer to the probe clamp. Make sure that the transducer is placed in the active port of the Vevo2100 system.
- (ii) Adjust your ultrasound imaging parameters (see ‘Setup of the Vevo Imaging Station’ in Equipment Setup) for both B-mode and nonlinear contrast mode imaging.
▲ CRITICAL STEP Ensure that all imaging parameters are kept the same, so data can be compared across animals.
- (iii) Warm the mouse for 5–10 min to dilate its blood vessels. Care should be taken to not overheat the animal.
! CAUTION The heat lamp can get very hot. Handle with care.
- (iv) After the animal has been warmed up, anesthetize for tail-vein cannulation. Mice are anesthetized using a table-top animal anesthesia system. Place the mouse in the induction chamber and open the valve of the medical air cylinder. Set the flow meter to 1.5 liters/min and the vaporizer to 5% isoflurane.
- (v) After 1–2 min, transfer the mouse to the handling table and re-route the anesthesia from the induction chamber to the nose cone. Fit the nose cone over the nose and mouth of the mouse. Pinch the toe to make sure that there is no reflex, and then lower the vaporizer to 2% for maintenance.
▲ CRITICAL STEP Use medical air instead of oxygen as the carrier for the isoflurane to increase the stability of the GVs.
- (vi) Fill a 1-ml syringe with a 27-gauge $\times$ 1/2 needle with 0.9% sodium chloride solution, and attach the end of the catheter tubing to the needle.
- (vii) Flush the catheter tubing with the 0.9% sodium chloride solution until the solution leaks from the catheter needle.
! CAUTION Failure to fill the tubing with liquid to remove the air from the catheter can lead to a gas embolism and death of the animal.
- (viii) Insert the catheter into the tail vein as described in Step 63A(v).
- (ix) Place eye ointment in both eyes of the mouse to prevent drying.
- (x) If desired, heart rate and respiratory rate monitoring can be accomplished as follows: place a small amount of electrode gel onto the 4 electrodes located at the corners of the mouse handling table, and secure each paw of the mouse to the electrodes with surgical tape. Make sure that the animal is placed in a supine position to allow for ultrasound probe access to the liver. Refer to **Figure 4d**.
- (xi) For body temperature monitoring, lubricate the temperature probe and insert gently into the rectum of the animal. Refer to **Figure 4d**.
- (xii) Apply ultrasound gel to the upper abdominal region of the mouse, right below the rib cage and place the transducer on top of the gel.
- (xiii) Starting with B-mode imaging, adjust the X, Y and Z positioning knobs to locate a slice of the liver where the inferior vena cava (i), portal vein (ii) and abdominal aorta (iii) can be observed in the same imaging plane. Once an imaging plane has been established, switch the imaging mode to nonlinear contrast.
- (xiv) Resuspend the GVs thoroughly by gently pipetting up and down, and then add 60 μL of solution (50 μL for injection + 10 μL to fill catheter tubing) to a 1.5-ml microcentrifuge tube. We typically use solutions with an $\text{OD}_{500,\text{ps}}$ value ~ 5 for Halo injections, although investigators can go higher or lower based on their experimental needs and design (injected concentrations with $\text{OD}_{500,\text{ps}}$ values as low as 1.5 can be detected in the IVC).
▲ CRITICAL STEP GVs must be mixed thoroughly each time before an injection; they are buoyant and will float to the solution surface very quickly. Make sure to inject immediately after remixing.
- (xv) Withdraw the GV solution into a .5-cc tuberculin syringe. With the needle pointing up, remove any bubbles by gently pulling the plunger down and up, and then secure into the syringe pump. Do not flick the syringe, as this will collapse GVs.

Box 1 | General signal considerations ● TIMING 5 min

In comparison with ^1H MRI, the concentration of Xe is ~ 5 orders of magnitude lower. This is compensated by its higher (~ 4 orders of magnitude) polarization. Hence, higher Xe partial pressure helps to increase the detectable signal, which is directly proportional to the xenon concentration in solution. The calculation to determine the Xe concentration in solutions can be done as follows using the Ostwald solubility coefficient:

Procedure

1. For the PBS solution, assume an Ostwald solubility coefficient similar to that of water: 0.11 liter/bar or 4.9 mM/bar (1 mole of an ideal gas corresponds to 22.41 liters).

The partial pressure of the Xe fraction (5%) for 1.2 bar of gas mixture in the system is then 0.05×1.2 bar, and the concentration of dissolved Xe is $[\text{Xe}] = 0.11 \text{ L/bar} \times 0.05 \times 1.2 \text{ bar} \times 4.9 \text{ mM/bar} = 295 \text{ }\mu\text{M}$ (keep in mind that only 26.4% is the NMR-active isotope ^{129}Xe).

2. To get an idea of how long it takes to saturate the solution with Xe, consider the following: it has been shown that bubbling water with a 2% (vol/vol) Xe mix at a flow corresponding to 100 SMLM (calibrated to N_2 ; i.e., ~ 140 SMLM for a 2% (vol/vol) Xe mix) exponentially saturates the solution with Xe with a time constant of ~ 8 s (see supplementary information of ref. 23). A 5% (vol/vol) Xe mix at 40 SMLM (described in this protocol) delivers $\sim 29\%$ less Xe per unit of time, and the time constant increases accordingly.

(xvi) Imaging under nonlinear contrast mode, acquire ultrasound images before, during and after a bolus injection of GVs, to observe the change in signal from the inflow of gas vesicles. A high-power burst sequence can also be applied to collapse any gas vesicles in the imaging plane and diminish the signal.

? TROUBLESHOOTING

(xvii) Steps 63B(xiii–xvi) can be repeated for other tissue regions of interest.

(xviii) At the end of the imaging session, turn off the isoflurane and remove the mouse from the nose cone. Wait for the animal to completely recover from the anesthesia before returning it back to the cage.

Production of hyperpolarized Xe and initial NMR calibrations ● TIMING 1 h

64| Preparation of GV samples for MRI: dilute the stock solution of GVs in PBS solution to achieve an $\text{OD}_{500, \text{PS}}$ value of ~ 0.33 . For each sample, 2 ml of GVs are needed.

▲ CRITICAL STEP Although Ana and Mega GVs are stable and survive multiple xenon re-deliveries and data sets, Halo GVs are very fragile. Therefore, use a fresh Halo sample for each measurement.

OD measurements should be performed before and after NMR measurements to ensure that GV concentrations stay constant.

65| *Startup of the polarizer (Steps 65–72: timing = 20 min).* Turn on the hyperpolarizer according to the manufacturer's protocols.

66| Select an absolute gas pressure in the system of ~ 1.2 bar (i.e., 0.2 bar above atmospheric pressure). This pressure is adjusted through the Daq/board-regulated pressure controller behind the sample and will be reached once the hyperpolarizer outlet valve has been opened. The manual pressure regulator directly at the gas tank should be set to a higher value (1.5 to 2 bar).

▲ CRITICAL STEP As the GVs are pressure-sensitive, an excessive pressure set point for the sample could later damage the GVs in solution when performing Step 77. On the other hand, the gas pressure determines the amount of dissolved Xe and eventually the image signal intensity. Considerations regarding the expected spin concentrations are described in **Box 1**.

67| Connect the gas-delivery tubing (coming from the hyperpolarizer) with the gas outlet tubing (going into the pressure controller) using glass connecting tube shown in **Figure 5b**.

68| Slowly open the outlet valve of the hyperpolarizer to pressurize the gas manifold and to achieve the selected total pressure of 1.2 bar in the entire system.

! CAUTION Backflow of air (oxygen) or humidity into the system and the Rb-containing optical pumping cell in the polarizer setup must be avoided at all times to prevent violent reactions with the reactive Rb.

69| Select a total gas flow rate of 250 SMLM through the system for the flow controller between the gas tank and the hyperpolarizer. The pressure controller will now maintain the 1.2 bar while releasing incoming excess gas at the selected flow rate.

PROTOCOL

- 70| Wait until the process temperatures have reached a steady state.
- 71| Stop the gas flow by setting the flow controller set point to 0 SMLM and isolate the hyperpolarizer by closing its outlet valve.
- 72| Depressurize the gas manifold by carefully unscrewing the glass tube connector while keeping the polarizer on standby.
- 73| *Checking of the gas-delivery conditions (Steps 73–82: timing = 10 min).* Decide whether the glass phantom should be used with one compartment for a single sample solution or with two compartments for including a second solution (e.g., a reference buffer solution without GVs); in the latter case, insert a 5-mm NMR tube into the glass phantom.
- 74| Load the sample solution(s) into the glass phantom and screw the capillary holder onto the inlet thread (distribute the capillaries as needed between the two compartments in case the 5-mm tube has been inserted). Make sure that the filling level is not too high, so that the NMR coil can also detect some of the gas atmosphere on top of the solution.
- 75| Connect the gas-delivery tube from the hyperpolarizer setup to the capillary holder of the phantom and the gas outlet tube to the other chem thread of the glass phantom.
- 76| Select a total flow rate through the polarizer of 250 SMLM and activate the flow. This will ensure that gas is flowing into the system once the next step is performed.
- 77| Carefully pressurize the PFA tubing system by slowly opening the outlet valve of the hyperpolarizer to reach the over-pressure of 0.2 bar.
- ▲ **CRITICAL STEP** Make sure that the shortcut valve is always open for pressurizing or depressurizing the sample; this avoids abrupt pressurization that could push the sample solution out of the glass phantom into the PFA tubing, which can later contaminate the flow or pressure controller.
- ▲ **CRITICAL STEP** When pressuring the GV-containing sample with a xenon gas mix, a fast change in pressure must be avoided and a rate of 0.1 bar/30 s should not be exceeded to ensure that the GVs remain intact.
- 78| Once the pressure has stabilized, ensure that the total flow rate of 250 SMLM through the polarizer is still active while the shortcut valve is open; this flow will now go through the bypass without bubbling the sample.
- 79| To bubble the sample, adjust the sample delivery flow to 40 SMLM.
- 80| For activating the flow, the pulse sequence must be executed. However, all radio frequency (RF) pulses should be turned off, as there is no sample in the NMR coil that absorbs the transmitted power. To do so, select the pulse-acquire FID sequence including the TTL triggers and set the amplitude of the excitation pulse to 0.
- 81| Choose a bubble time of ~10–20 s and a waiting time after bubbling of ~1–2 s.
- 82| Set the number of repetitions to 5–10 and start the sequence while watching the sample for sufficient gas dispersion during periodic gas delivery and possible unwanted excessive foam formation inside the glass phantom. Ensure that any foam and gas bubbles disappear within the waiting time after bubbling (typically 1–2 s).
- ▲ **CRITICAL STEP** Prolonged bubbling could collapse GVs via shear forces. The flow of gas through the sample should be long enough to have an adequate xenon signal but short enough to avoid stressing the GVs too much. The subsequent waiting time should be as short as possible to minimize the longitudinal relaxation of the xenon but long enough to ensure that all bubbles collapse.
- ? **TROUBLESHOOTING**
- 83| *Preparative MRI measurements (Steps 83–91: timing = 10 min).* Insert the pressurized phantom carefully into the NMR probe that is already mounted inside the magnet in its correct position.
- 84| If desired, adjust the temperature of the sample with the VTU of the spectrometer. The sample data from **Figure 6** were acquired at room temperature ($T = 298\text{ K}$).
- ▲ **CRITICAL STEP** If another set point, e.g., body temperature, is chosen, the sample might need 20–40 min to reach stable temperature. It is possible to continue with the next steps, but Step 87 should not be performed until stable conditions are reached.

Box 2 | Flip-angle calibration for hyperpolarized xenon ● TIMING 10 min

Flip-angle calibration is usually done with a series of repetitive excitations and subsequent analysis of the obtained signal intensities. It is used to derive pulse amplitudes for the shaped excitation pulses, as well as to determine the saturation power in μT . As hyperpolarization does not self-replenish, it is convenient to use a constant small flip-angle excitation and analyze the signal decay after a single Xe delivery as follows:

Additional materials

- Nonselective NMR pulse sequence for acquiring a series of Xe solution signals after a single Xe delivery. This is implemented by using a simple (pulse–acquire–crusher) sequence with fast repetition between the excitations. As an example, the sequence can have a 100- μs block excitation pulse of adjustable amplitude, followed by a 9.72-ms FID sampling time (99 data points collected) and crusher gradients to remove all remaining transverse magnetization.
- Evaluation tool such as a gnuplot (<https://sourceforge.net/projects/gnuplot/>) fitting routine or Origin (<https://www.origin.com/usa/en-us/store/download>) to fit the data to a user-defined exponential decay model function.

Procedure

1. Choose the above-mentioned nonselective NMR sequence with single Xe delivery and multiple acquisitions.
2. Select a relatively small excitation pulse amplitude (assuming $\alpha \sim 7^\circ\text{--}11^\circ$).
- ▲ **CRITICAL STEP** If the amplitude is too high, the signal decay will be either too fast or not monotonous; if the amplitude is too low, the observed signal will be too noisy over the entire series.
3. Run the sequence with a repetition time of 18 ms and NEX = 250 repetitions; the crusher gradients burn for ~ 1 ms at an amplitude of ~ 90 G/cm.
4. The incoming time domain signal will show a series of 250 truncated FID signals. The spectrometer software may display this as a series of 250 Fourier transform (FT) magnitude spectra with decaying signal intensity (**Fig. 6f,g**). For an evaluation without custom-made scripts such as a gnuplot fitting routine, perform the following steps.
5. Export the magnitude spectra data as a .txt file and load them into Origin.
6. Perform ParaVision standard postprocessing: each spectrum contains 128 data points, of which only the peak heights are important.
7. Plot the concatenated spectra and evaluate the envelope formed by the peak heights. This envelope can be reconstructed with the '↓Analysis ↓Signal Processing ↓Envelope' tool in Origin using ~ 100 smoothing points.
8. Fit the envelope using the function $I(x) = M_0 + I_0 \sin \alpha (\cos \alpha)^{(x/p - 1)}$ with α as the desired unknown flip angle, M_0 as a constant offset representing the nonvanishing noise level for large x , and p as the fixed number of displayed data points between two peaks. The fit should be performed for the range where the actual data is still above the noise level (**Fig. 6h,i**). This evaluation neglects T_1 relaxation during the short acquisition time. Fitting the observed signal intensity to this function therefore yields the flip angle α that the 100- μs block pulse achieved for the selected pulse amplitude.
9. The scaling factor for the pulse amplitude of a 10-fold-longer standard block pulse (1 ms) is then given by $90^\circ / (10 \times \alpha)$. Use this for the reference gain for all derived pulse amplitudes.
10. Knowing the amplitude for a 90° pulse of 1-ms length, derive the pulse power in μT as follows: a full nutation takes 4 ms, and hence the precession about the B_1 field occurs at 250 Hz. The gyromagnetic ratio for ^{129}Xe is $\gamma = 11.777$ kHz/mT. Hence, the applied field strength is $B_1 = 250/11,777$ mT = 0.0212 mT = 21.23 μT . This can be used to derive the pulse powers for a desired saturation level in μT .

85| Tune and match the ^1H and ^{129}Xe resonators, perform a global automatic shim on the sample and acquire a proton reference image with automated adjustments for ^1H resonance frequency, reference attenuation (flip angle) and receiver gain according to the spectrometer manufacturer's protocol.

86| Set the Xe delivery flow to 40 SLM, 10-s bubbling time and 2-s wait time or as optimized in Step 82.

87| Take a 1D Xe NMR spectrum with the pulse-acquire FID sequence including the TTL triggers with an ~ 350 p.p.m. bandwidth and four signal averages. If the amplitude for a flip angle of $\sim 90^\circ$ is not known, proceed with the steps in **Box 2** (and **Fig. 6f–i**) to determine the reference pulse.

88| Perform Fourier transform and automatic phase correction to display the NMR spectrum. Find the resonance frequency of the left intense peak (**Fig. 6a**) and use it as the transmitter frequency for further ^{129}Xe MRI acquisition.

? TROUBLESHOOTING

89| For the imaging experiments to follow, enter the reference pulse power into the pulse sequence control interface (typically the pulse amplitude for a 1-ms block pulse to achieve a 90° flip of the magnetization). Switch to an MRI sequence and choose the same bubbling time, wait time and transmitter frequency.

90| Ensure that the excitation for either the EPI or the RARE sequence is effectively a 90° pulse and that other sequence parameters are set as in the equipment preparation section. The number of acquisitions should be set to one.

91 Run the Xe MRI acquisition without any saturation pulses (deactivate the magnetization transfer (MT) module) and without a phase-encoding gradient (GSP in Bruker's ParaVision), and set the receiver gain such that the maximum incoming signal reaches ~30% of the full analog-to-digital converter (ADC) range.

Hyper-CEST MRI

92 For Hyper-CEST MRI acquisition, follow the steps in option A for acquiring an entire image series with different saturation offsets for obtaining spectral information of the CEST response (e.g., **Fig. 6b**) or option B for acquiring data with only one on-resonant and one off-resonant saturation to obtain a single Hyper-CEST image (e.g., **Fig. 6c–e**). Spectral information is obtained when changing the frequency of the saturation pulse. This results in a so-called z-spectrum whose characteristic shape is influenced by xenon's exchange properties with the GV. Option A should be performed if the exact resonance frequency of Xe inside the GVs is not known. Images with improved quality can then be obtained subsequently with option B.

(A) Imaging series and z-spectrum ● TIMING 45 min

(i) Select the desired number of frequency offsets and enter the values for the offsets. The example data in **Figure 6b** were taken with 31 offsets covering a range from –300 p.p.m. to +175 p.p.m. with an increment of ~16 p.p.m. (ii) Select the saturation power B_1 and saturation time t_{sat} . The example here used saturation conditions of $B_1 = 15 \mu\text{T}$, and $t_{\text{sat}} = 5 \text{ s}$.

! CAUTION Ensure that the power limits for the NMR coil are not exceeded, as this might permanently damage the coil.

▲ CRITICAL STEP It is not useful to choose a saturation time that exceeds the T_1 relaxation time of Xe in water; the signal loss is then dominated by intrinsic relaxation and not by the CEST effect²⁴.

▲ CRITICAL STEP The choice of the right saturation conditions is critical to optimal z-spectra quality and for correct interpretation of saturation transfer in xenon MR images. As a rule of thumb, short but strong saturation is desired first. If multiple agents (different GV types in this context) are present, then the saturation strength should be reduced and the saturation time should be adapted similarly. (iii) Start the image series acquisition and make sure that the system does not perform any auto-adjustments that overwrite any of the previous settings (GOP command in Bruker's ParaVision).

▲ CRITICAL STEP If the images are acquired individually because the operating system does not allow an array of measurements as in Bruker's ParaVision, it is important that all images be taken with the same receiver gain and that the system does not automatically readjust the receiver gain before each acquisition.

(iv) Open the image evaluation tool of the spectrometer software and draw a region of interest (ROI) for signal integration in the xenon MR images that were acquired for different saturation pulse frequencies.

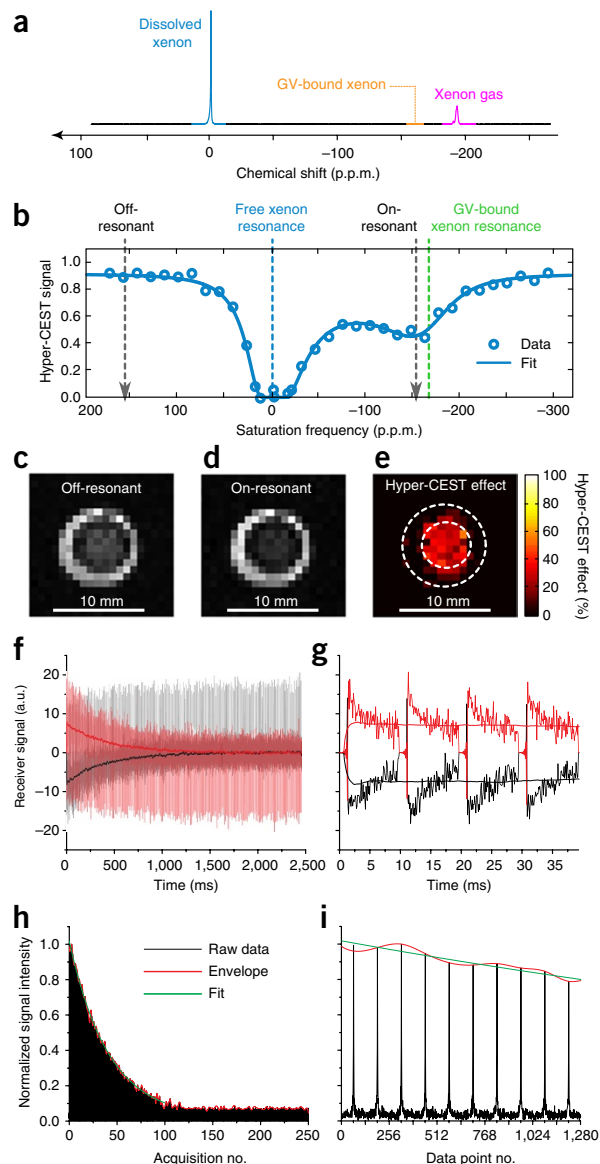


Figure 6 | ^{129}Xe HyperCEST MRI. (a) 1D ^{129}Xe NMR spectrum of a solution containing GVs, illustrating the signal from Xe gas in the space above the sample solution (magenta), the below-the-noise signal from GV-bound Xe (yellow) and the signal from the dissolved Xe in solution (cyan), which is assigned to 0 p.p.m. for referencing. (b) Hyper-CEST z-spectrum of Mega GVs (0.67 nM in PBS, $T = 295 \text{ K}$). On-resonant (–156.5 p.p.m.) and off-resonant (+156.5 p.p.m.) saturation ($B_1 = 15 \mu\text{T}$, $t_{\text{sat}} = 5 \text{ s}$) produced different signal intensities, and the spectrum was fitted to the exponential of two Lorentzian line shapes^{28,29}. Analysis based on Bloch-McConnell equations predicts that the GV-bound Xe resonance frequency dashed (green line) is ~170 p.p.m. (c,d) Axial off- (c) and on-resonant (d) saturation xenon MR images (16 times averaged). A 5-mm NMR tube containing Mega GVs is nested inside a 10-mm NMR tube containing only PBS. (e) The differential images are normalized by the intensity of the off-resonant image, demonstrating the ~50% Hyper-CEST contrast of GVs. The white dashed circles indicate the glass walls of the double-tube phantom. (f–i) Flip-angle calibration. (f) Full time-domain FID signal train of 250 acquisitions (black: real domain; red: imaginary) showing an overall exponential decay (bold solid lines) that is regularly interrupted by the ringing artifacts of opening the ADC gate. (g) Zoom-in of the first four FIDs in f. (h) Concatenated magnitude spectra after Fourier transformation of the data displayed in f. The fitting (green line) is based on the first 100 spectra, giving $\alpha = 12.6^\circ$. (i) Zoom-in of the first 10 spectra in h.

(v) Plot the signal intensity of the ROI versus the saturation frequency offset to obtain the z-spectrum.

? TROUBLESHOOTING

(vi) Analyze the obtained data: the full signal loss for a 0 p.p.m. saturation offset is the direct saturation of free Xe in solution. The GV-related response is expected at an offset ~ 170 to -190 p.p.m. This local minimum in signal intensity corresponds to the frequency for the best Hyper-CEST response. Optimum conditions for option B are then obtained for choosing this offset for on-resonant saturation and its negative value for the off-resonant control image.

▲ **CRITICAL STEP** Although the true GV-bound xenon resonance frequency in the sample data is at ~ 170 p.p.m.

(green dashed line, **Fig. 6b**), the maximum xenon depolarization spectrally appears at a different frequency, which is shifted toward that of free xenon (blue dashed line) at ~ 156.5 p.p.m. This difference is caused by so-called spillover between the GV-bound xenon resonance and that of free xenon. This would not occur if the Hyper-CEST signal between both resonances would recover entirely back to the baseline. Two parameters affect the 'intensity' of spillover: (i) the exchange regime (given by the ratio of the xenon exchange rate and chemical shift difference in Hz) and (ii) the strength of the saturation pulse B_1 . To avoid spillover effects, one can use GVs with slower Xe exchange or reduce the saturation power.

(vii) An accurate evaluation is based on normalized z-spectra that also account for the T_1 relaxation. For this purpose, acquire data points with no saturation (i.e., setting the saturation time to 0) and negligible relaxation: set the number of saturation offsets to 5, the saturation frequency offset to +300 p.p.m. and—most importantly—the saturation power and duration to 0 μ T and 0 s, respectively. Start the acquisition and draw an ROI in the xenon MR images for signal integration.

(viii) Plot the signal intensity of the ROI for the 5 acquisitions; they should be almost identical and the average value can be used for normalizing the values of the z-spectrum obtained in Step 92A(v).

(B) Obtaining of a single Hyper-CEST image from two saturation frequencies ● TIMING 20 min

- (i) Set the number of saturation offsets to 2.
- (ii) Enter the on-resonant saturation frequency offset and the off-resonant saturation frequency offset according to the z-spectrum interpretation in Step 92A(vi)).
- (iii) Select the saturation power B_1 and saturation time t_{sat} . The example in **Figure 6e** used saturation conditions of $B_1 = 15 \mu$ T, and $t_{\text{sat}} = 5$ s.
 ! **CAUTION** Ensure that the power limits for the NMR coil are not exceeded, as this might permanently damage the coil.
 ▲ **CRITICAL STEP** It is not useful to choose a saturation time that exceeds the T_1 relaxation time of Xe in water²⁴; the signal loss is then dominated by intrinsic relaxation and not by the CEST effect.
- (iv) Enter the desired number of image averages, i.e., the number of excitations (NEX), to achieve good signal intensity – e.g., NEX = 16.
- (v) Start the sequence that will perform 2× NEX acquisitions.
- (vi) Check the results in the image display software.
- (vii) The difference image displaying the areas with the Hyper-CEST response can be obtained by subtracting the image with on-resonant saturation from the image with off-resonant saturation using the image display software; if desired, normalize the difference image by the off-resonant control image.
 ▲ **CRITICAL STEP** Ensure that the intensity scaling is the same for the two images when performing any difference image calculations.

Shutting down of the system ● TIMING 10 min

93| Set the flow into the polarizer to 0 SMLM.

94| Close the polarizer gas outlet and shut down the system according to the manufacturer's instructions.

95| Ensure that the shortcut valve is open.

96| Carefully depressurize the phantom by slowly ramping down the set point of the pressure controller.

▲ **CRITICAL STEP** Fast depressurizing may damage the GVs and render them unusable for future experiments or quality control.

97| Disconnect the sample tube from the PFA tubings and store them for later experiments.

TABLE 5 | Troubleshooting table.

Steps	Problem	Possible reason	Solution
1A(iv), 1A(vi) 1B(iii)	Cultures are growing too slowly	Not enough viable cells	Increase the amount of starter inoculum

(continued)

TABLE 5 | Troubleshooting table (continued).

Steps	Problem	Possible reason	Solution
	Cell death	Contamination Incubator conditions (aeration, pH, temperature and lighting) are not optimal	Start fresh from primary cultures and follow sterile procedures Loosen the cap/foil to allow aeration, and monitor CO ₂ levels. For Ana cultures, ensure that the temperature inside the incubator does not increase during light cycle owing to heat from the lamps, and confirm proper exposure of the cultures to light
1A(vii)–(xi), 1B(iv)–(vi)	Poor GV yield	Cultures are not confluent	Allow cultures to grow for a few more days. Confluency is determined by the color and turbidity of the cultures as described in the procedure and shown in Figure 2
1A(xi), 1B(vi)		Incomplete lysis	Dilute the concentrated cell suspension with lysis buffer and/or allow lysis to proceed longer
1A(vii)–(xi), 1 B(iv)–(vi)		Collapse of GVs	Make sure that the cultures are not subjected to agitation >100 r.p.m., and handle flasks gently while transporting and placing them on hard surfaces
1A(x), 1B(vi)		Inefficient retrieval of buoyant cells from the separatory funnel	Thoroughly wash the inner walls of the funnel with media/buffer to retrieve the buoyant cells sticking to the sides (especially for Halo)
1A(xii)	Incomplete separation of GVs from lysate/subnatant	Lysate is too dense, requiring longer spins for GVs to float to the top	Increase the time of centrifugation and/or dilute the lysate with PBS before spin
1A(xii), 1C(v)	Collapse of GVs during spins	Hydrostatic pressure on GVs is greater than critical collapse pressure	Prepare aliquots of suspensions in more tubes to reduce the height of the column (volume in a single Falcon tube should not exceed 40 ml)
1A(xv), 1B(vii)		Pressurization of tubes while opening and closing	Ensure that microcentrifuge tubes are closed very gently. The caps of such tubes should not make a snap sound when closed
8	NHS-amine reaction did not work	NHS will hydrolyze in the presence of water/moisture	Ensure that the DMSO used for storage is anhydrous Limit exposure to ambient moisture Store it in a desiccated environment
4–10	GVs aggregate after reaction with NHS-moiety	Conjugation of moieties on the GVs destabilizes the protein Presence of surfactants in the reactant sample	Conjugate fewer moieties on the GV surface by reducing the concentration of NHS-moiety and/or reaction time Purify the NHS-moieties using chromatography or other methods
9, 10	Subsequent reaction using NHS-moiety-conjugated GVs does not work	Presence of excess NHS-moieties in solution	Repeat dialysis until excess NHS-moieties are below the level of detection
11–19	Incomplete GvpC stripping	GV concentration is too high for efficient removal of GvpC	Dilute native Ana GVs in PBS to an OD of ~5 before adding 10 M urea buffer
20–38	Low or no GvpC yield	Low expression of GvpC variant	Check plasmid for mutations, prepare fresh transformations and try reducing temperature and increasing duration of expression
		Poor binding to the Ni column	Ensure that the column is charged and increase incubation time with resin or add more resin
		Inefficient retrieval of inclusion bodies	Ensure that inclusion pellets are not lost during repeated rounds of centrifugation and resuspension
39–41	Incomplete GvpC re-addition	GV concentration is too high for re-addition reaction to proceed to completion	Reduce GV OD ₅₀₀ value during GvpC re-addition. Confirm re-addition by SDS-PAGE and Coomassie staining
		GvpC concentration is too low	Inaccurate quantification of GvpC eluate

(continued)

TABLE 5 | Troubleshooting table (continued).

Steps	Problem	Possible reason	Solution
39–41	Protein precipitation is seen after re-addition	Excess unbound GvpC forms precipitates in PBS	Reduce stoichiometric excess factor of GvpC during re-addition
42–45	No fluorescence from SpyCatcher-GFP-labeled GVs	Incomplete SpyTag functionalization	Confirm GvpC-SpyTag re-addition using SDS–PAGE
		Incomplete SpyTag-SpyCatcher reaction	Increase excess factor of SpyCatcher during reaction and/or increase the reaction time
46A	Collapse curves are inconsistent or noisy	Incomplete collapse	Make sure that there are no air leaks in the connectors or bubbles in the sample
		Presence of detergents and/or incomplete GvpC addition	Always run control GV samples such as Δ GvpC and GvpC _{WT} to ensure that the re-addition was done under the proper conditions and corroborate with SDS–PAGE results
46C	Too much black background on TEM grid	GV sample contains contaminants or phosphate-containing molecules	Use dialysis or centrifugally assisted purification to resuspend GVs in 10 mM HEPES with 150 mM NaCl buffer
		Uranyl acetate precipitate is present in solution	Avoid extended exposure to ambient light. Uranyl acetate will precipitate when exposed to light and UV
	Too many or too few GVs on the TEM grid	Incorrect GV OD measurement	Measure GV OD value and double-check the dilution calculation
		GVs were not given adequate time to adhere to charged grids.	Excessive wicking can cause too many GVs to come off grids. Reduce how much solution is being wicked off
47–62	Only the top of the GV sample for <i>in vitro</i> ultrasound phantoms shows contrast	The concentration of GVs is too high	Lower the concentration of GVs
47–62	The signal from GVs and polystyrene beads is very different	The concentration of GVs or polystyrene beads is too high or too low	Match the GV concentration to the concentration of beads
63A, 63B	Weak or no signal after injection	IVC is not in the image	Move the transducer to find the optimal location in the IVC and try to align the imaging target with the natural focus of the transducer
		GVs collapsed before or during injection	Check that there is no pressure when injecting through the catheter by manually injecting a small volume of saline with a 30-gauge syringe. If you feel the pressure, re-adjust the catheter into the tail vein and ensure that there are no blockages, allowing the saline to flow smoothly
		Transmit power is set too high	Ensure that transmit power does not generate an acoustic pressure on the GVs that exceeds the critical collapse pressure
82	Excessive foam formation	The overall protein concentration in the sample is too high	Reduce the protein concentration or add an antifoam agent such as pluronic L81 (~0.1% by volume); however, such an agent can also have an impact on the integrity of the protein structure
88	The Xe NMR spectrum does not show any signal or shows very distorted signals	The excitation pulse is too weak	Increase the amplitude of the excitation pulse
		The field homogeneity is poor	Perform another shim
		There are still gas bubbles in the sample while acquiring the FID signal	Increase the waiting time after bubbling, or consider adding an antifoam agent
		The Rb is oxidized and no hyperpolarized Xe is produced	Check the laser absorption profile of the hyperpolarizer and replace the Rb in the pumping cell if necessary

(continued)

TABLE 5 | Troubleshooting table (continued).

Steps	Problem	Possible reason	Solution
92A(v)	No CEST response from the GVs is observed in the z-spectrum, or the image with on-resonant saturation does not show a true signal loss	GVs have collapsed GV concentration is too low Saturation power is too low or saturation time is too short	Measure the OD value of the sample and prepare a fresh sample from stock solution if necessary Add GVs from stock solution Increase the saturation power or the saturation time
92A(v)	The saturation response from the GVs is too broad and merges into the direct saturation response	Saturation power is too high	Reduce the saturation power

98 | Turn off the VTU.

? TROUBLESHOOTING

Troubleshooting advice can be found in **Table 5**.

● TIMING

GV production and purification

Media preparation: 1 h

Step 1A(i–iv), growing of Ana starter cultures: 2 weeks

Step 1A(v,vi), subculture of Ana for GV production: 2 weeks

Step 1A(vii–xi), harvesting of Ana GVs: 1 d

Step 1A(xii–xvii), isolation and purification of GVs from Ana cultures: 2–3 d

Step 1B(i–iii), growing of Halo cultures: 2 weeks

Step 1B(iv–vi), harvesting of Halo GVs: 1 week

Step 1B(vii–xi), isolation and purification of GVs from Halo cultures: 2 d

Step 1C(i–iv), heterologous expression of Mega GVs in *E. coli*: 2 d

Step 1C(v–x), harvesting and purification of Mega GVs: 2 d

GV quantification

Step 2, measurement of pressure-sensitive OD using NanoDrop instrument: 15 min

Step 3, measurement of GV protein concentration: 15 min

Chemical functionalization of GVs

Steps 4–10, chemical functionalization of GVs: 2 d (includes 1 d for dialysis)

Steps 11–19, (optional) preparation of Δ GvpC GVs: 12 h

Steps 20–38, preparation of recombinant GvpC: 1 d

Steps 39–41, preparation of GVs with recombinant GvpC: 1 d

Steps 42–45, functionalization of SpyCatcher-functionalized GVs: 5 h

GV characterization

Step 46A, determination of critical collapse with pressurized absorbance spectroscopy: 40 min

Step 46B, GV preparation for DLS: 5 min

Step 46C, preparation of GV samples for TEM: 30 min

Ultrasound imaging of GVs *in vitro*

Steps 47–55, agarose phantom preparation and sample loading: 3 h

Steps 56–62, phantom imaging: 1 h (can take longer, depending on the number of samples and mode of acquisition)

Ultrasound imaging of GVs *in vivo*

Step 63A: GV imaging *in vivo* using the Verasonics ultrasound system: 45 min

Step 63B: Halo GV imaging *in vivo* using the Vevo ultrasound system: 45 min (timing may vary based on number of acquisitions)

Production of hyperpolarized Xe and initial NMR calibrations

Steps 64–91, preparation of GV sample and MRI equipment for data acquisition: 1 h

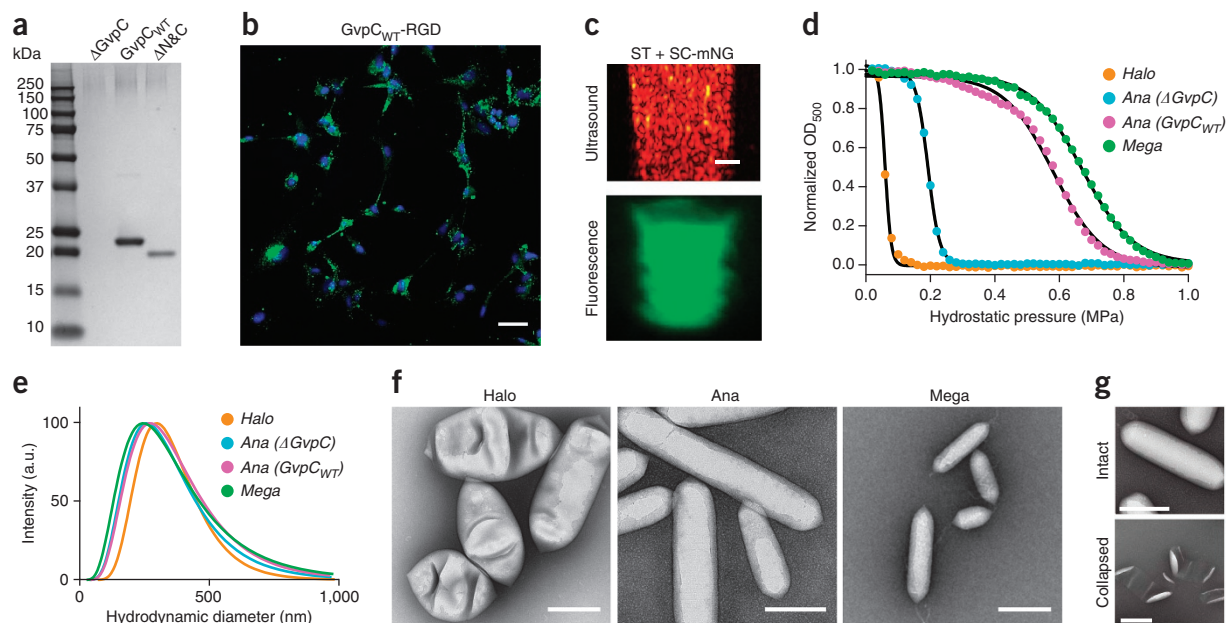


Figure 7 | Anticipated results for GV characterization and *in vitro* ultrasound imaging. **(a)** Coomassie-stained SDS-PAGE gel indicating complete removal of native GvpC (Δ GvpC, lane 2) and re-addition of two recombinant GvpC variants: GvpC_{WT} and Δ N&C GvpC (lanes 3 and 4). The smeared bands ~250 kDa represent GvpA polymers forming the inner GV scaffold, whereas GvpC bands are visible at lower molecular weights. **(b)** Confocal fluorescence image showing dual-functionalized Ana GV with RGD peptide and Alexa Fluor-488 (green) after a 24-h incubation with U87 glioblastoma cells (DAPI-stained nuclei, blue). Scale bar, 50 μ m. **(c)** Dual-mode imaging of GV functionalized with SpyCatcher-mNeonGreen with ultrasound (top panel) and fluorescence (excitation = 506 nm, emission = 550 nm) (bottom panel). Scale bar, 1 mm. **(d)** Collapse pressure profiles for GV variants (Halo, Mega, Ana Δ GvpC and Ana GvpC_{WT}). **(e)** DLS measurements showing typical size distributions for purified Halo, Mega, Δ GvpC and GvpC_{WT} Ana GV. **(f)** Representative TEM images for purified Halo, Ana and Mega GV. Scale bars, 200 nm. **(g)** TEM showing intact and collapsed Ana GV. Scale bars, 200 nm **(f–g)**. More information on the exact gene and protein sequence can be found by accessing the plasmid information using the Addgene ID given in the Materials section. a.u., arbitrary units; Δ N&C GvpC, GvpC variant without the N- and C-terminal regions of the protein. **a–c** adapted with permission from ref. 14, American Chemical Society. **g** adapted with permission from ref. 8, Nature Publishing Group.

Step 92A, imaging series and z-spectrum: one experimental session: 45 min

Step 92B, single Hyper-CEST image from two saturation frequencies: one experimental session: 20 min

Steps 93–98, shutting down of the system: 10 min

Box 1, general signal considerations: 5 min

Box 2, flip-angle calibration for hyperpolarized xenon: 10 min

ANTICIPATED RESULTS

GV expression and purification

Ana and Halo GV expression and purification. Healthy, viable Ana and Halo cultures look dark green (**Fig. 2c**) and light pink (**Fig. 2d**), respectively, upon reaching confluency. **Figure 2e,f** shows the separation of buoyant Ana and Halo cells from spent medium. For Halo cultures, the buoyant cells are visible as a thin layer or ring of milky pink at the top of the liquid phase. After lysis, the lysate will appear milky because of the presence of GV. After one round of centrifugally assisted floatation, the GV will appear as a white layer at the top with a green (Ana) or pink (Halo) liquid subnatant, along with a pellet of cell debris at the bottom. For the Halo, a mixed layer of white (Halo GV) and milky pink (unlysed Halo cells) may be visible after the initial spins. **Figure 2g,h** shows representative images of the purified Ana and Halo GV in 2-ml tubes after the final spin, with the white GV forming a dense layer at the top and a clear subnatant of PBS. **Figure 2j** shows tubes containing milky white solutions of purified Ana and Halo GV ($\sim$ OD_{500,ps} = 6) resuspended in PBS. The typical GV yield is $\sim$ 3 ml of OD_{500,ps} = 10 for Ana and Halo GV per flask of culture.

Mega GV expression and purification in *E. coli*. Although native expression of Ana and Halo GV usually results in the majority of the cells being buoyant, most of the *E. coli* cells heterologously expressing Mega GV reside in the pellet after centrifugation. In our experience, the typical yield of Mega GV is $\sim$ 1.5 mg/l cells per unit of OD₆₀₀. **Figure 2i** shows a representative image of purified and unclustered Mega GV forming dense white layer at the top, with a clear subnatant of PBS. **Figure 2j** shows a milky white solution of purified and unclustered Mega GV resuspended in PBS ($\sim$ OD_{500,ps} = 6).

GV functionalization

Ana GvpC is purified from inclusion bodies in *E. coli*. The recombinant GvpC can be harvested with yields of up to 40 mg protein per liter of culture. The inclusion bodies yield >95% pure protein (**Fig. 7a**). GvpC-based functionalization relies upon complete removal of the native Ana GvpC layer before reconstitution with recombinant protein. Urea treatment results in nearly complete (>95%) removal of native GvpC, indicated by the absence of GvpC on SDS-PAGE (**Fig. 7a**) and a lowered critical collapse pressure (shown in the following section on GV characterization). GVs retain their milky white appearance throughout the functionalization process. The binding of recombinant GvpC variants to GVs can be assayed by SDS-PAGE and Coomassie staining. Δ GvpC Ana GVs do not exhibit a band ~25 kDa (molecular weight of GvpC), whereas wild-type and truncated GvpC variants bind to GVs and exhibit a distinct band corresponding to their theoretical molecular weight (**Fig. 7a**).

Figure 7b shows an application using dual-functionalized GVs for targeted cell-specific imaging. In this example, Ana GVs functionalized with GvpC fused to peptide RGD are chemically conjugated to Alexa Fluor 488. The functionalized GVs bind to integrin receptors on glioblastoma cells via the RGD motif and the fluorescent labeling of GVs enables rapid assessment of targeting efficacy using confocal microscopy. SpyTag-GvpC binds to GVs with similar productivity and stoichiometry as wild-type GvpC. We have previously shown that each modified GV has an average of 1,000 SpyTag functionalities¹⁴. The collapse pressure of SpyTag GVs is not altered upon SpyCatcher binding, indicating that attachment of a large 37.9-kDa protein to the GV surface does not adversely affect GvpC strengthening of the GV wall¹⁴.

Functionalization of SpyTag-GVs with SpyCatcher-GFP enables multimodal imaging of GVs with ultrasound and fluorescence. Ultrasound images were acquired in an agarose phantom with a Verasonics L22-14V transducer at 19 MHz, 5.0 V and F number 3. The ultrasound images show similar echogenicity between fluorescently labeled GVs and wild-type and unreacted controls (**Fig. 7c**). Optical images taken through the green channel of a BioRad Chemidoc MP system indicate that SpyTag-SpyCatcher-GFP labeled GVs are fluorescent (**Fig. 7c**). Fluorescence quantification of Halo GVs after chemical conjugation to Alexa Fluor 488 indicates that functionalized GVs are ~40-fold more fluorescent than native GVs.

GV characterization

Critical collapse pressure of GVs. The collapse pressure profiles of purified Ana, Halo and Mega GVs, as well as those of functionalized GVs, are obtained by measuring change in the OD_{500nm} values of GV samples under increasing hydrostatic pressure. The mean collapse pressures are 59 kPa (Halo), 587 kPa (Ana GvpC_{WT}) and 750 kPa (Mega) (**Fig. 7d**). Ana GVs without GvpC have a substantially lower critical collapse pressure of 193 kPa. Collapse pressure measurements also indicate the degree of mechanical strengthening by recombinant GvpC variants.

DLS measurements. DLS values for hydrodynamic diameters exhibit variability due to the biogenic nature of GVs. Representative DLS profiles are given in **Figure 7e**. Hydrodynamic diameters of Halo GVs range from 260 to 320 nm (polydispersity = 0.15–0.21), those of Δ GvpC Ana GVs range from 250 to 360 nm (polydispersity = 0.21–0.24), those of GvpC_{WT} Ana GVs range from 240 to 340 nm (polydispersity = 0.17–0.26) and those of Mega GVs range from 200 to 380 nm (polydispersity = 0.23–0.34).

TEM visualization of GVs. GVs appear as bright particles on a dark background under negative staining with uranyl acetate. Typically, TEM allows visualization of the morphology, size, shape and overall surface features of GVs. **Figure 7f** shows representative TEM images for purified wild-type Halo, Ana and Mega GVs. Although Halo GVs appear spindle shaped, Ana and Mega GVs are more cylindrical, with conical tips in the longitudinal dimension. Although the size and shape of GVs are determined primarily by the genotype, each type possesses a certain degree of heterogeneity. For example, Ana GVs have length distribution with an s.d. of 35% of the mean⁶. It is also important to note that not all Halo GVs have a spindle morphology, and a small sub-population may appear more cylindrical with biconical ends²⁷. Ana GVs that are produced in their native host are usually longer and wider than heterologously expressed Mega GVs. Typical GV dimensions obtained from TEM images are summarized in **Table 4**.

The high resolution provided by TEM allows the visualization of ribs on the surface of the gas vesicles at higher magnifications. **Figure 7g** shows intact and collapsed Ana GVs, demonstrating that collapse causes complete rupture and opening of the protein shell, leading to a flattened pancake-like structure. GVs chemically functionalized with biotin can be coated with streptavidin and visualized by TEM⁸.

In vitro ultrasound imaging

Typical ultrasound images acquired in the fundamental mode and after second harmonic filtering show contrast from intact wild-type (WT) and Δ GvpC Ana GVs (**Fig. 8a**, top panel). Acoustic collapse of GVs leads to elimination of ultrasound signal (**Fig. 8a**, bottom panel), confirming that contrast arises from the intact GVs. Lowering of collapse pressure is typically associated with higher nonlinear signals from GV samples, allowing tuning of GV harmonic response by modification or removal of GvpC. The ultrasound contrast from Halo GVs and their utility for nonlinear imaging *in vitro* and *in vivo* has been demonstrated in our previous work⁸. The acoustic behavior of Halo GVs at ultrasound frequencies of 12.5–27.5 MHz has also been investigated through modeling and experiments, suggesting acoustic buckling as the mechanism underlying generation of nonlinear signals¹⁰. In parallel, we recently developed an amplitude modulation scheme taking advantage of the nonlinear pressure dependence of backscattered signals in engineered Ana GVs, allowing selective imaging of these nanostructures⁹.

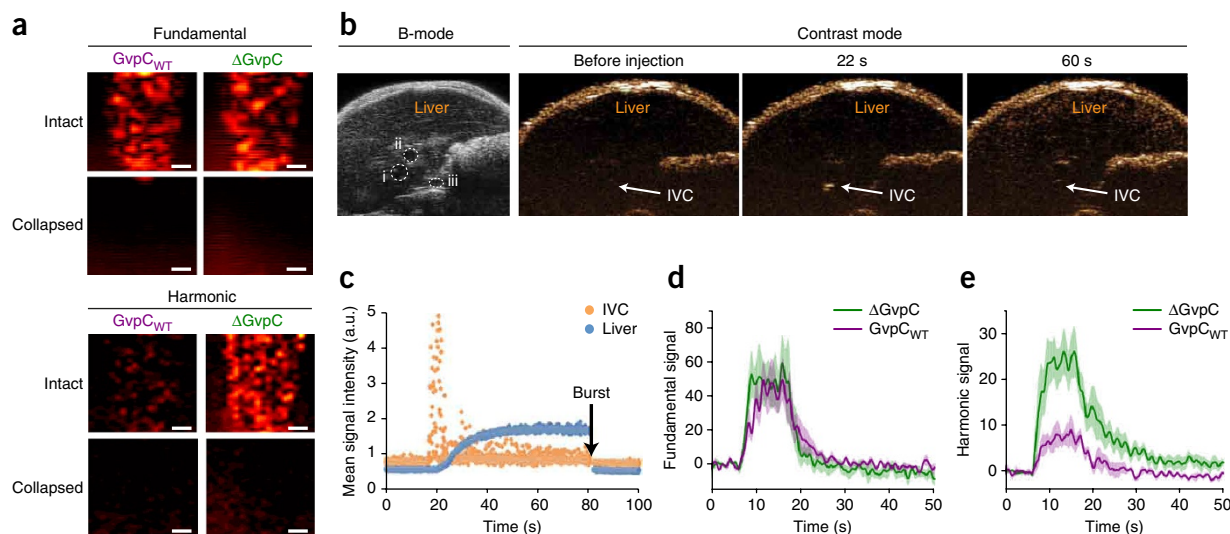


Figure 8 | Anticipated results for *in vitro* and *in vivo* ultrasound imaging. **(a)** Fundamental and second harmonic images for intact Δ GvpC and GvpC_{WT} Ana GV (top panel) acquired using the Verasonics L11-4v transducer with a 4.46-MHz transmission and band-pass-filtered $\sim$ 4.46 and 8.92 MHz, respectively. A high-power burst from the transducer is used to collapse the GV (bottom panel). Scale bars, 1 mm. **(b)** B-mode image of the liver showing the (i) inferior vena cava (IVC), (ii) portal vein and (iii) abdominal aorta. Contrast-mode images acquired using high-frequency ultrasound show the liver and IVC with the baseline contrast image before injection of Halo gas vesicles. Following a bolus injection of gas vesicles, contrast enhancement is seen in the IVC, followed by the liver. **(c)** Time intensity curve depicting the nonlinear contrast signal change over time for the IVC (orange) and the liver (blue) before, during and after an intravenous administration of Halo gas vesicles. **(d,e)** Time intensity curve in the **(d)** fundamental and **(e)** second harmonic mode depicting the change in acoustic signal in the IVC before, during and after steady infusion with Δ GvpC and GvpC_{WT} Ana GV. Shaded regions represent SEM ($N = 6$). All procedures involving animal use must be performed in accordance with institutional guidelines and regulations and approved by relevant animal care and use committees. All animal experiments reported in this article received approval from the institutional animal care and use committees of the California Institute of Technology or the Sunnybrook Research Institute. **a, d** and **e** adapted with permission from ref. 14, American Chemical Society.

In vivo ultrasound imaging

When imaging gas vesicles *in vivo*, we suggest starting with a vascularized organ such as the liver. A normal liver will look homogeneous under B-mode ultrasound, with vessel cross-sections appearing as hypoechoic regions (**Fig. 8b**, left panel). When imaging a mouse in the transverse plane, it is possible to locate a region of the liver containing the IVC.

After gas vesicles are administered intravenously via the lateral tail vein, nonlinear contrast enhancement will be observed first in the IVC, followed by the liver (**Fig. 8b**). Using the VevoLAB software, regions of interest (ROIs) encompassing the IVC and liver can be drawn, and a time intensity curve of the mean contrast signal change over time within each ROI can be plotted.

As shown in **Figure 8c**, once a 50- μ l bolus of gas vesicles has been injected, an initial bulk signal is observed in the IVC, which will decrease over time as the gas vesicles are dispersed into the circulation. Signal enhancement in the liver will follow a few seconds after with a steady signal increase and then a plateau. The signal plateau in the liver is probably due to the uptake of nonfunctionalized gas vesicles by the Kupffer cells, similar to that for many other nano-sized particles. This contrast signal can then be eliminated by using a high power burst that will collapse all the gas vesicles in the imaging plane.

It is possible to engineer Ana GV to produce enhanced nonlinear signals *in vivo* (**Fig. 8d,e**). After bolus injection of GV, ultrasound imaging of the IVC is performed in fundamental and harmonic modes for up to 1 min (transmission at 4.46 MHz and reception filtered $\sim$ 4.46- and 8.9-MHz center frequencies, respectively). The mean signal intensity plotted as a function of time (**Fig. 8d,e**) shows that Δ GvpC Ana GV produce higher harmonic signals as compared with Ana GvpC_{WT} *in vivo*, whereas their fundamental signals are comparable. This observation is also consistent with the *in vitro* data in **Figure 8a**.

Hyperpolarized Xe MRI imaging of GV

Typical GV concentrations in the nanomolar to picomolar range will not allow direct observation of the GV-bound Xe. A ^{129}Xe NMR spectrum therefore shows only the signal from dissolved xenon gas (assigned to 0 p.p.m.) and the resonance from Xe gas on top of the solution (appearing $\sim$ 190 p.p.m. to the right of the solution peak) (**Fig. 6a**). Data obtained in Step 92A(v) should look similar to the spectrum in **Figure 6b**, except that it is not yet normalized. Taking the relaxation and saturation-free reference signal from Step 92A(viii) into account, the spectrum should yield intensities between 0 and 1, similar to **Figure 6b**. It should be noted that the longer the saturation time, the more the baseline has dropped below 1.

Data from Step 92B can be displayed as the (normalized) Hyper-CEST effect map, which gives a signal intensity as a percentage. The original data are plotted in **Figure 6c,d** and the Hyper-CEST effect map in **Figure 6e**. This Hyper-CEST effect map is obtained by postprocessing the off- and on-resonant xenon MR images as follows: Hyper-CEST effect map = (off-resonant – on-resonant)/off-resonant × 100.

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

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- Ferrara, K., Pollard, R. & Borden, M. Ultrasound microbubble contrast agents: fundamentals and application to gene and drug delivery. *Annu. Rev. Biomed. Eng.* **9**, 415–447 (2007).
- Kaufmann, B.A. & Lindner, J.R. Molecular imaging with targeted contrast ultrasound. *Curr. Opin. Biotechnol.* **18**, 11–16 (2007).
- Weissleder, R. *et al.* Ultrasmall superparamagnetic iron oxide: characterization of a new class of contrast agents for MR imaging. *Radiology* **175**, 489–493 (1990).
- Lee, J.-H. *et al.* Artificially engineered magnetic nanoparticles for ultra-sensitive molecular imaging. *Nat. Med.* **13**, 95–99 (2007).
- Caravan, P., Ellison, J.J., McMurry, T.J. & Lauffer, R.B. Gadolinium (III) chelates as MRI contrast agents: structure, dynamics, and applications. *Chem. Rev.* **99**, 2293–2352 (1999).
- Walsby, A.E. Gas vesicles. *Microbiol. Rev.* **58**, 94–144 (1994).
- Pfeifer, F. Distribution, formation and regulation of gas vesicles. *Nat. Rev. Microbiol.* **10**, 705–715 (2012).
- Shapiro, M.G. *et al.* Biogenic gas nanostructures as ultrasonic molecular reporters. *Nat. Nanotechnol.* **9**, 311–316 (2014).

- Maresca, D. *et al.* Nonlinear ultrasound imaging of nanoscale acoustic biomolecules. *Appl. Phys. Lett.* **110**, 073704 (2017).
- Cherin, E. *et al.* Acoustic behavior of *Halobacterium salinarum* gas vesicles in the high-frequency range: experiments and modeling. *Ultrasound Med. Biol.* **43**, 1016–1030 (2017).
- Shapiro, M.G. *et al.* Genetically encoded reporters for hyperpolarized xenon magnetic resonance imaging. *Nat. Chem.* **6**, 629–634 (2014).
- Hane, F.T. *et al.* *In vivo* detection of cucurbit[6]uril, a hyperpolarized xenon contrast agent for a xenon magnetic resonance imaging biosensor. *Sci. Rep.* **7**, 41027 (2017).
- Barskiy, D.A. *et al.* NMR hyperpolarization techniques of gases. *Chem. Eur. J.* **23**, 725–751 (2017).
- Lakshmanan, A. *et al.* Molecular engineering of acoustic protein nanostructures. *ACS Nano* **10**, 7314–7322 (2016).
- Li, N. & Cannon, M.C. Gas vesicle genes identified in *Bacillus megaterium* and functional expression in *Escherichia coli*. *J. Bacteriol.* **180**, 2450–2458 (1998).
- Gilad, A.A. & Shapiro, M.G. Molecular imaging in synthetic biology, and synthetic biology in molecular imaging. *Mol. Imaging Biol.* **19**, 373–378 (2017).
- Zakeri, B. *et al.* Peptide tag forming a rapid covalent bond to a protein, through engineering a bacterial adhesin. *PNAS* **109**, E690–E697 (2012).
- Schröder, L. Chapter 8 HyperCEST Imaging. In *Chemical Exchange Saturation Transfer Imaging: Advances and Applications*. pp 121–158 (Pan Stanford Publishing, 2017).
- Schröder, L., Chapter 17 - Xenon Biosensor HyperCEST MRI A2 - Albert, Mitchell S. In *Hyperpolarized xenon for NMR and MRI* (ed. Hane, F.T.) pp 263–277 (Academic Press, Boston, 2017).
- Witte, C. *et al.* Hyperpolarized xenon for NMR and MRI applications. *JoVE* **67**, E4268 (2012).
- Witte, C., Kunth, M., Rossella, F. & Schröder, L. Observing and preventing rubidium runaway in a direct-infusion xenon-spin hyperpolarizer optimized for high-resolution hyper-CEST (chemical exchange saturation transfer using hyperpolarized nuclei) NMR. *J. Chem. Phys.* **140**, 084203 (2014).
- Nikolaou, P. *et al.* Near-unity nuclear polarization with an open-source ¹²⁹Xe hyperpolarizer for NMR and MRI. *PNAS* **110**, 14150–14155 (2013).
- Kunth, M., Witte, C., Hennig, A. & Schröder, L. Identification, classification, and signal amplification capabilities of high-turnover gas binding hosts in ultra-sensitive NMR. *Chem. Sci.* **6**, 6069–6075 (2015).
- Kunth, M., Witte, C. & Schröder, L. Continuous-wave saturation considerations for efficient xenon depolarization. *NMR Biomed.* **28**, 601–606 (2015).
- Smith, R. & Peat, A. Growth and gas-vacuole development in vegetative cells of *Anabaena flos-aquae*. *Arch. Microbiol.* **58**, 117–126 (1967).
- Buchholz, B., Hayes, P. & Walsby, A. The distribution of the outer gas vesicle protein, GvpC, on the *Anabaena* gas vesicle, and its ratio to GvpA. *Microbiology* **139**, 2353–2363 (1993).
- Simon, R.D. Morphology and protein composition of gas vesicles from wild type and gas vacuole defective strains of *Halobacterium salinarum* strain 5. *Microbiology* **125**, 103–111 (1981).
- Kunth, M., Witte, C. & Schröder, L. Quantitative chemical exchange saturation transfer with hyperpolarized nuclei (qHyper-CEST): sensing xenon-host exchange dynamics and binding affinities by NMR. *J. Chem. Phys.* **141**, 194202 (2014).
- Zaiss, M., Schnurr, M. & Bachert, P. Analytical solution for the depolarization of hyperpolarized nuclei by chemical exchange saturation transfer between free and encapsulated xenon (HyperCEST). *J. Chem. Phys.* **136**, 144106 (2012).